

# Bayesian inference methods for ecology

## Theory

What is Bayesian inference?

Probability 101

Bayes' theorem in biology

Priors and likelihood

Interpretation of posteriors

Markov Chain Monte Carlo

## Applications

Population dynamics in  
microscopic and  
macroscopic communities

Phylogeny and model  
selection

ODE generative models:  
Morphogen patterning of  
embryonic tissues

# Example 1.1: Determining a proportion

## Hypothesis testing What about p-values?

(AKA, the dangers of p-values, and how Bayesian inference comes to the rescue)

Question: Is proliferation more probable than cell death?

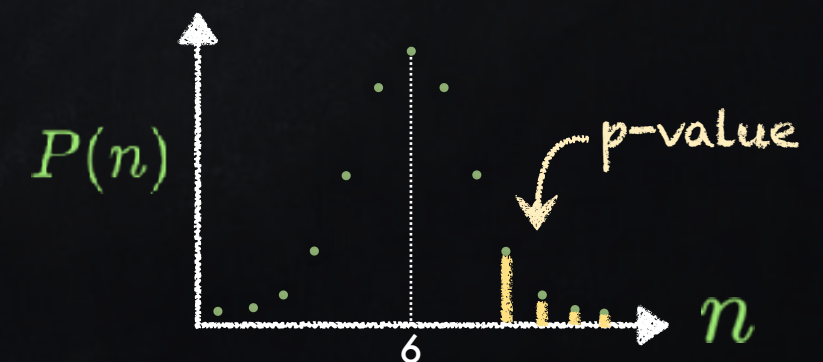
### P-value strategy:

Set null hypothesis: Rate proliferation same as death  $p = 1/2$

Perform experiment: Measure  $N = 12$  events and count number of proliferations  
For instance in this case you measured  $n = 9$  proliferations

Calculate probability of observation: Binomial distribution  $P(n) = \binom{N}{n} p^n (1-p)^{N-n}$

calculate p-value:  
(deviation from null) Probability the observation or more extreme values



$$P(n=9) + P(n=10) + P(n=11) + P(n=12) \simeq 0.073$$

Evaluate significance: P-value is  $> 0.05$  therefore we can't reject the null hypothesis

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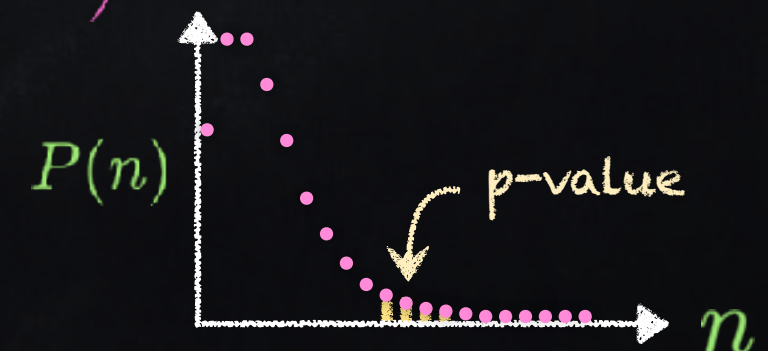
### P-value strategy:

Set null hypothesis: Rate proliferation same as death  $p = 1/2$

Perform experiment: Observe cells until  $d = 3$  deaths are detected, for instance  $n = 9$  proliferations were observed before  $d = 3$  (different intention but same data as previous example!)

Calculate probability of observation: Negative binomial  $P(n) = \binom{n+d-1}{n} p^n (1-p)^d$

calculate p-value: (deviation from null) Probability the observation or more extreme values



$$P(n=9) + P(n=10) + P(n=11) + \dots = 0.033$$

Evaluate significance: P-value is  $< 0.05$  therefore we can reject the null hypothesis

# Example 1.1: Determining a proportion

## Hypothesis testing What about p-values?

(AKA, the dangers of p-values, and how Bayesian inference comes to the rescue)

Question: Is proliferation more probable than cell death?

### Bayesian strategy:

~~Set null hypothesis:~~

Perform experiment:

Measure  $N = 12$  events  
and count number of  
proliferations  $n = 9$

Observe cells until  $d = 3$  deaths  
 $n = 9$  proliferations were observed

Calculate likelihood  
of observation:

Binomial distribution

$$P(n|p) = \binom{N}{n} p^n (1-p)^{N-n}$$

Negative binomial

$$P(n|p) = \binom{n+d-1}{n} p^n (1-p)^d$$

Calculate posterior  
distribution of  $p$

Uninformative prior

$$P(p|n=9) \propto P(n=9|p) = C p^9 (1-p)^3$$

Posterior is independent of the observatory intention (as it should be)

Calculate HDR:

$$\text{HDR}(95\%) = [0.091, 0.537]$$

Contains 0.5 so  $p=0.5$  is  
a credible value!

# Example 1.1: Determining a proportion

## Hypothesis testing What about p-values?

(AKA, the dangers of p-values, and how Bayesian inference comes to the rescue)

Question: Is proliferation more probable than cell death?

Bayesian strategy:

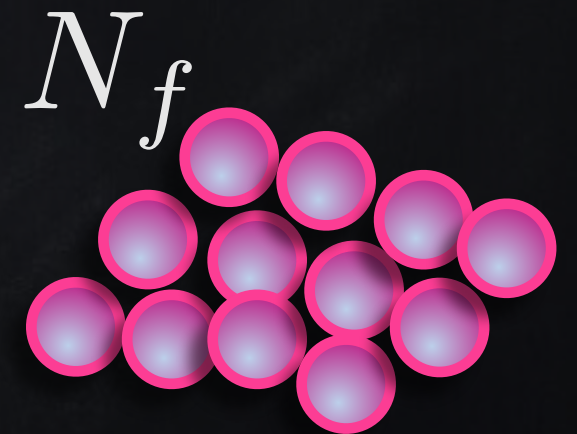
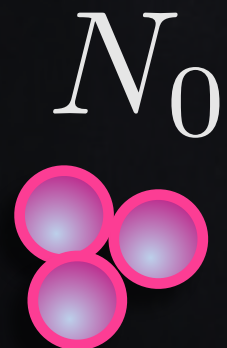
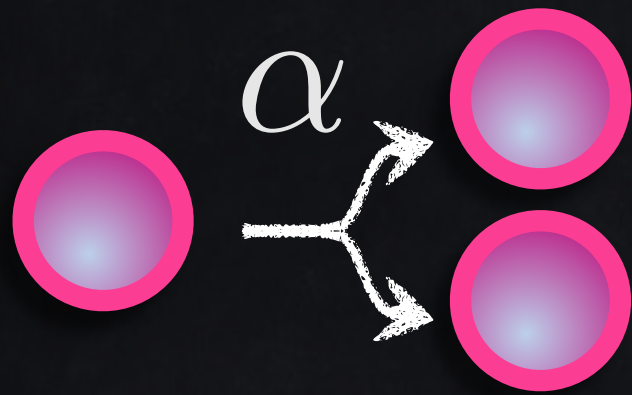


Bayesian inference only depends on the data,  
not on the observer intention



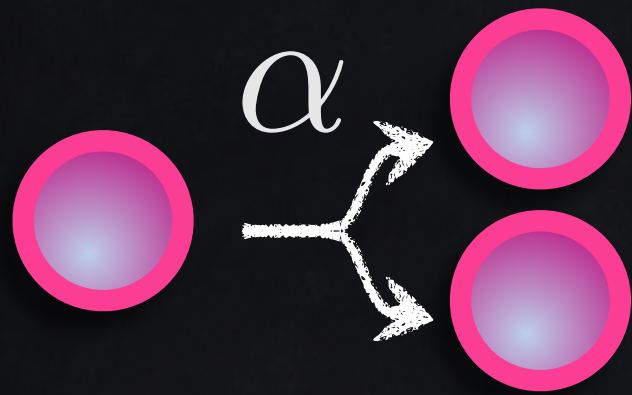
Hypothesis testing does not affect the inference process,  
this allows for multiple hypothesis testing

# Example 1.2: Population growth



How do I infer the values of  $\alpha$  and  $\delta$ ?

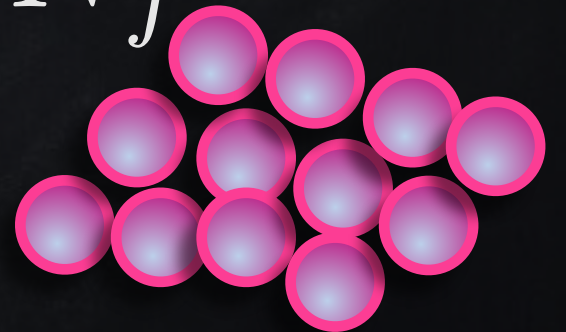
# Example 1.2: Population growth



$N_0$

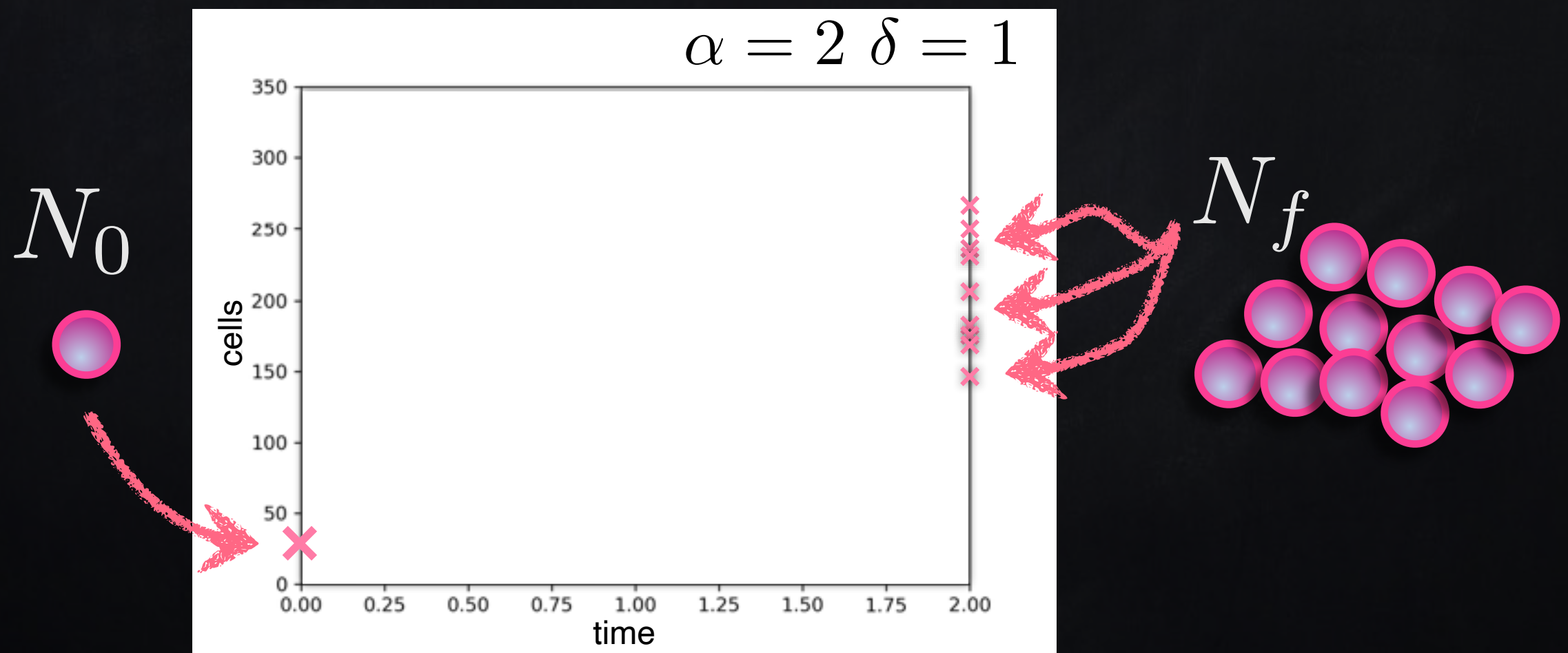
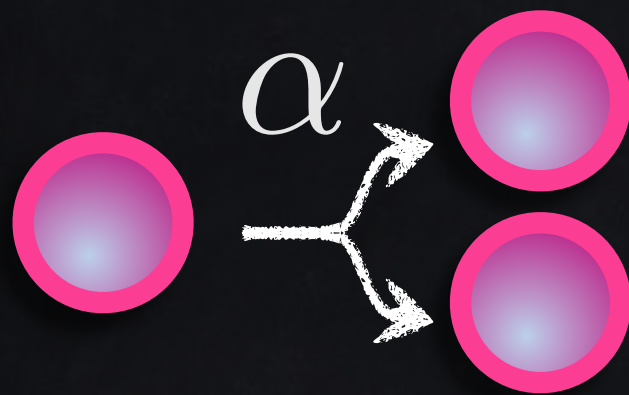


$N_f$



How do I infer the values of  $\alpha$  and  $\delta$ ?

# Example 1.2: Population growth

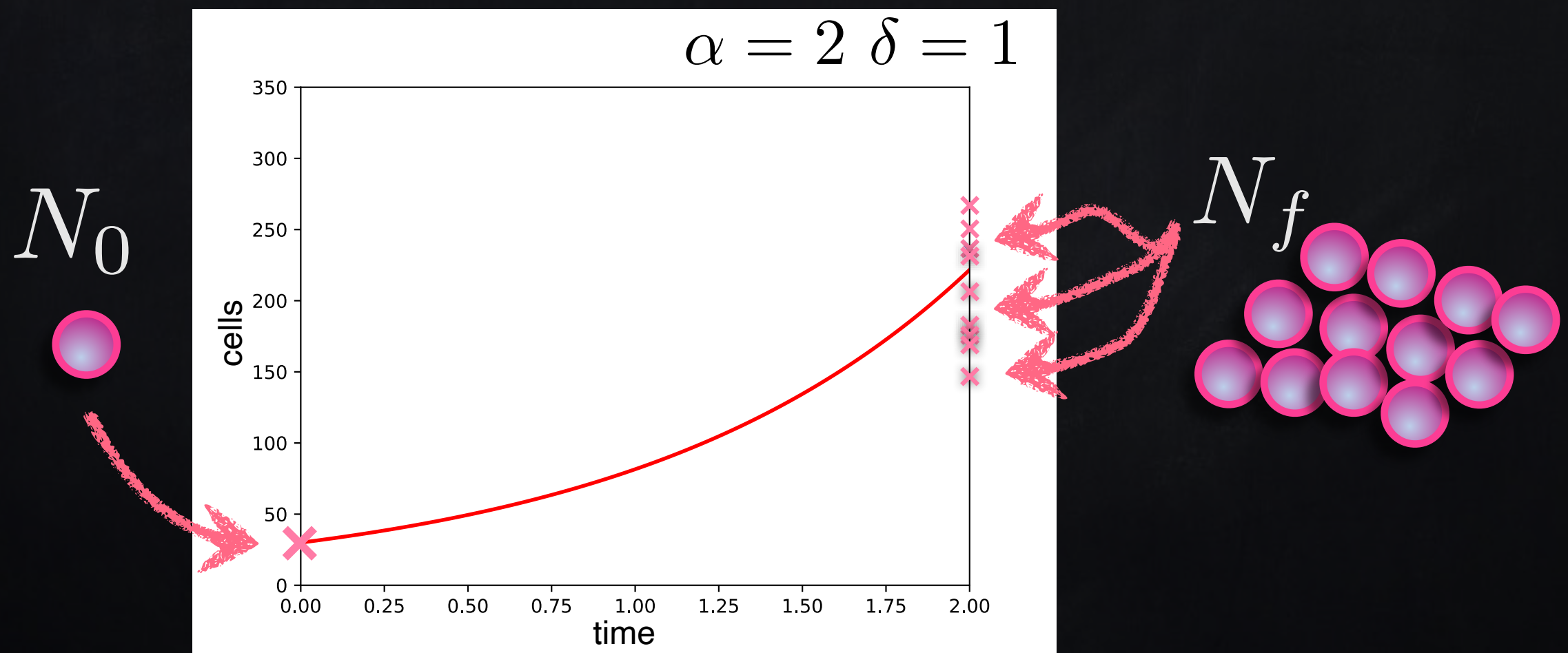


How do I infer the values of  $\alpha$  and  $\delta$ ?

# Example 1.2: Population growth



$$\frac{dN(t)}{dt} = \alpha N - \delta N$$



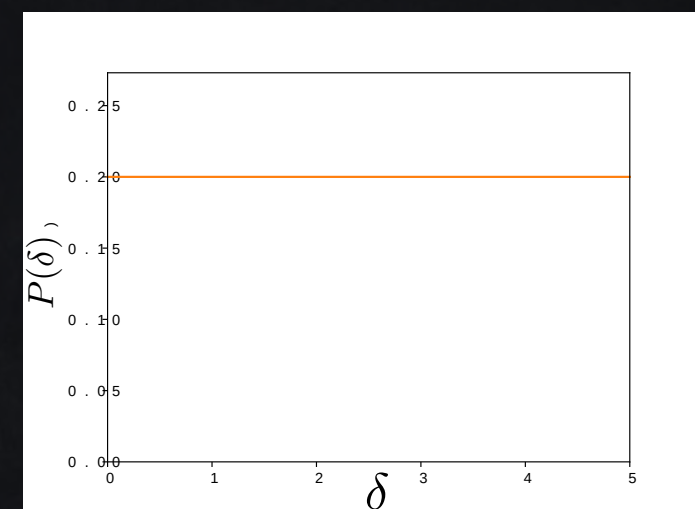
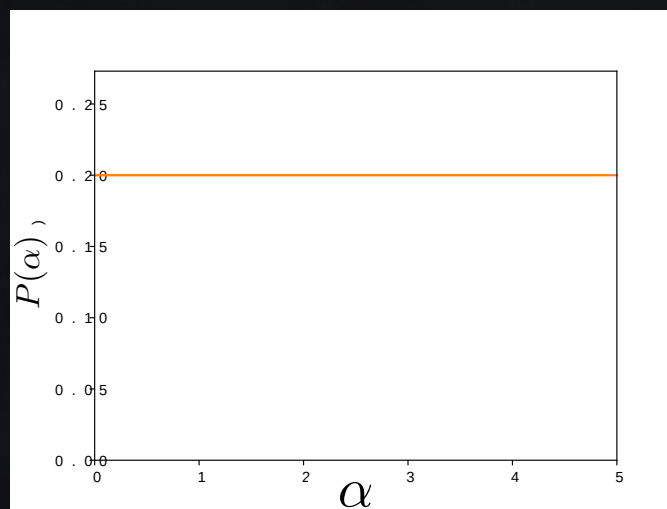
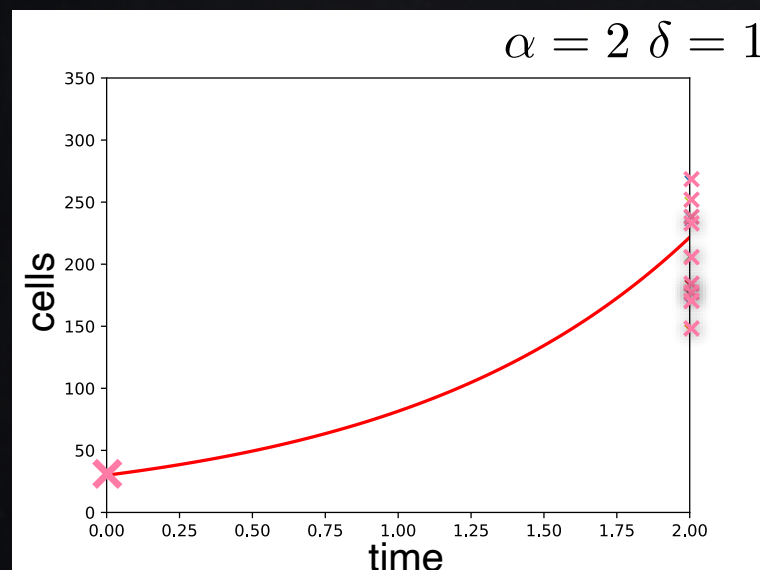
How do I infer the values of  $\alpha$  and  $\delta$ ?

# Example 1.2: Population growth



$$\frac{dN(t)}{dt} = \alpha N - \delta N$$

## Prior distributions

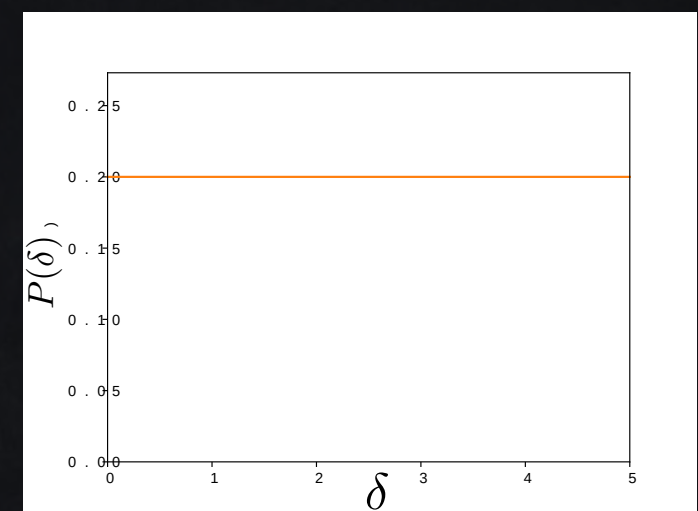
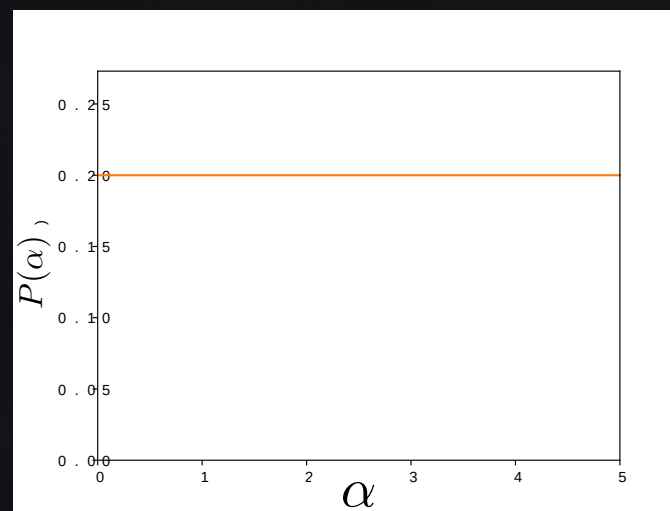
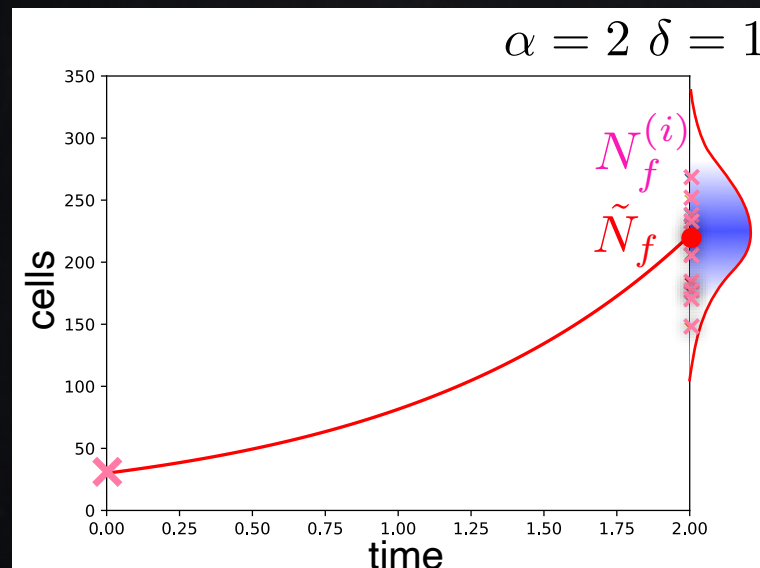


# Example 1.2: Population growth



$$\frac{dN(t)}{dt} = \alpha N - \delta N$$

## Prior distributions



## Likelihood

$$P(\text{data}|\alpha, \delta) = \prod_{i=1}^N P(\text{datapoint } i|\alpha, \delta) = \prod_{i=1}^N \mathcal{N}(\tilde{N}_f - \tilde{N}_f^{(i)}, \sigma)$$

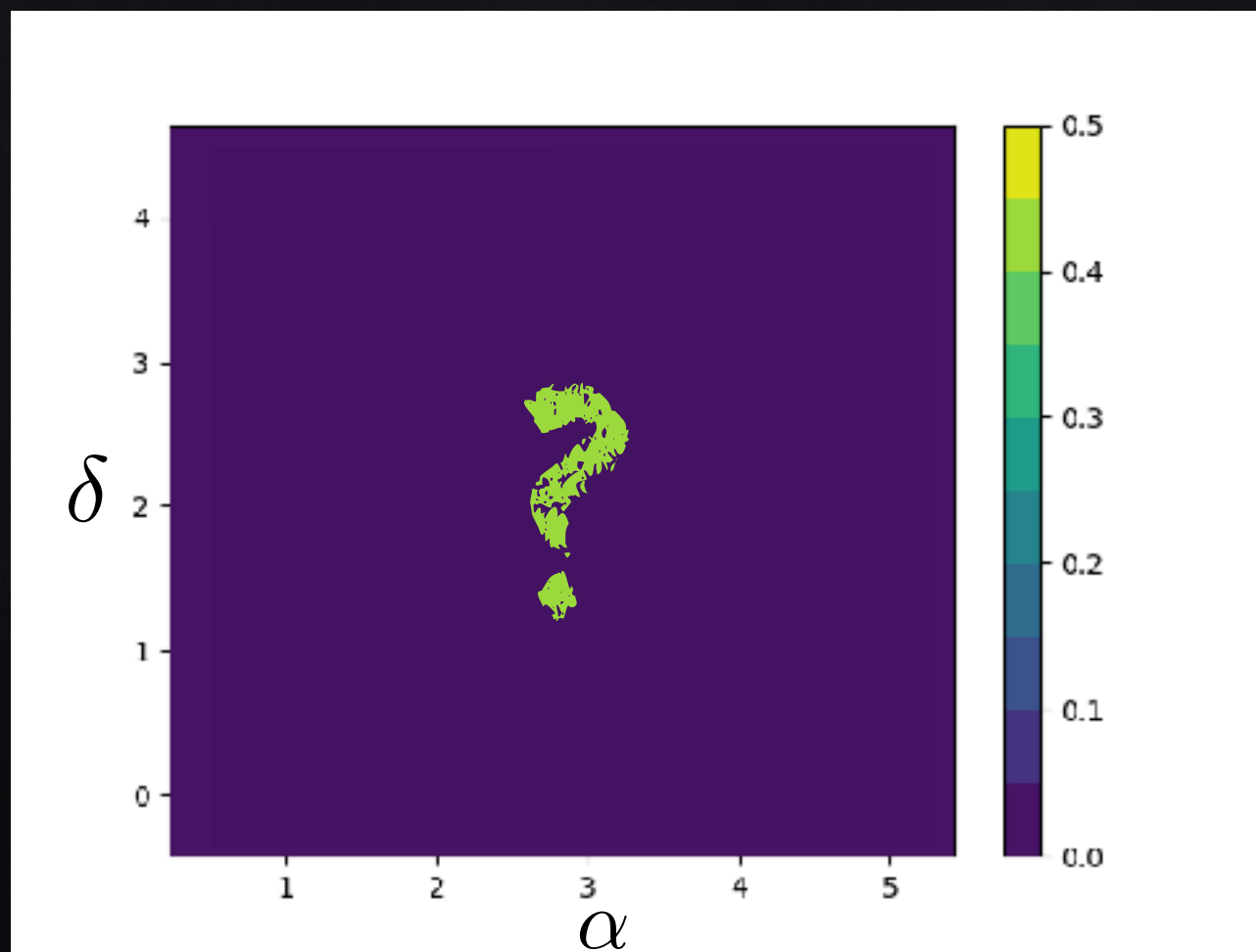
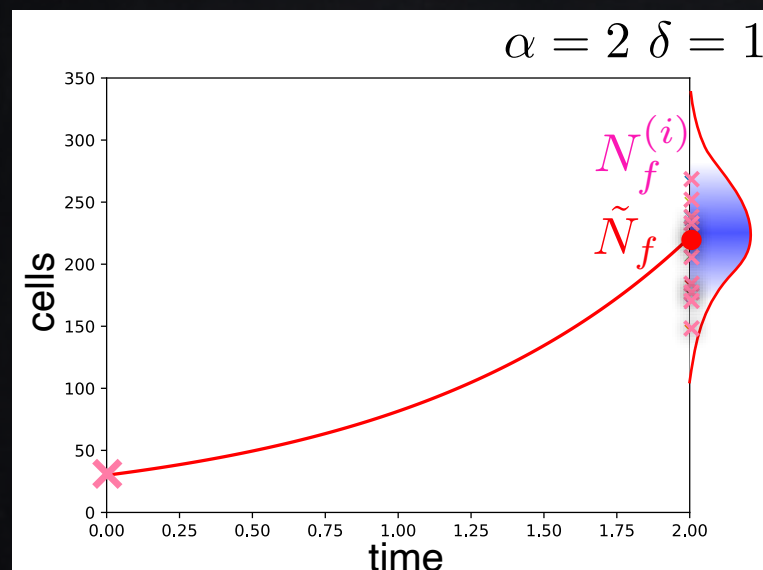
## approximate likelihood distance

# Example 1.2: Population growth



$$\frac{dN(t)}{dt} = \alpha N - \delta N$$

## Posterior distributions

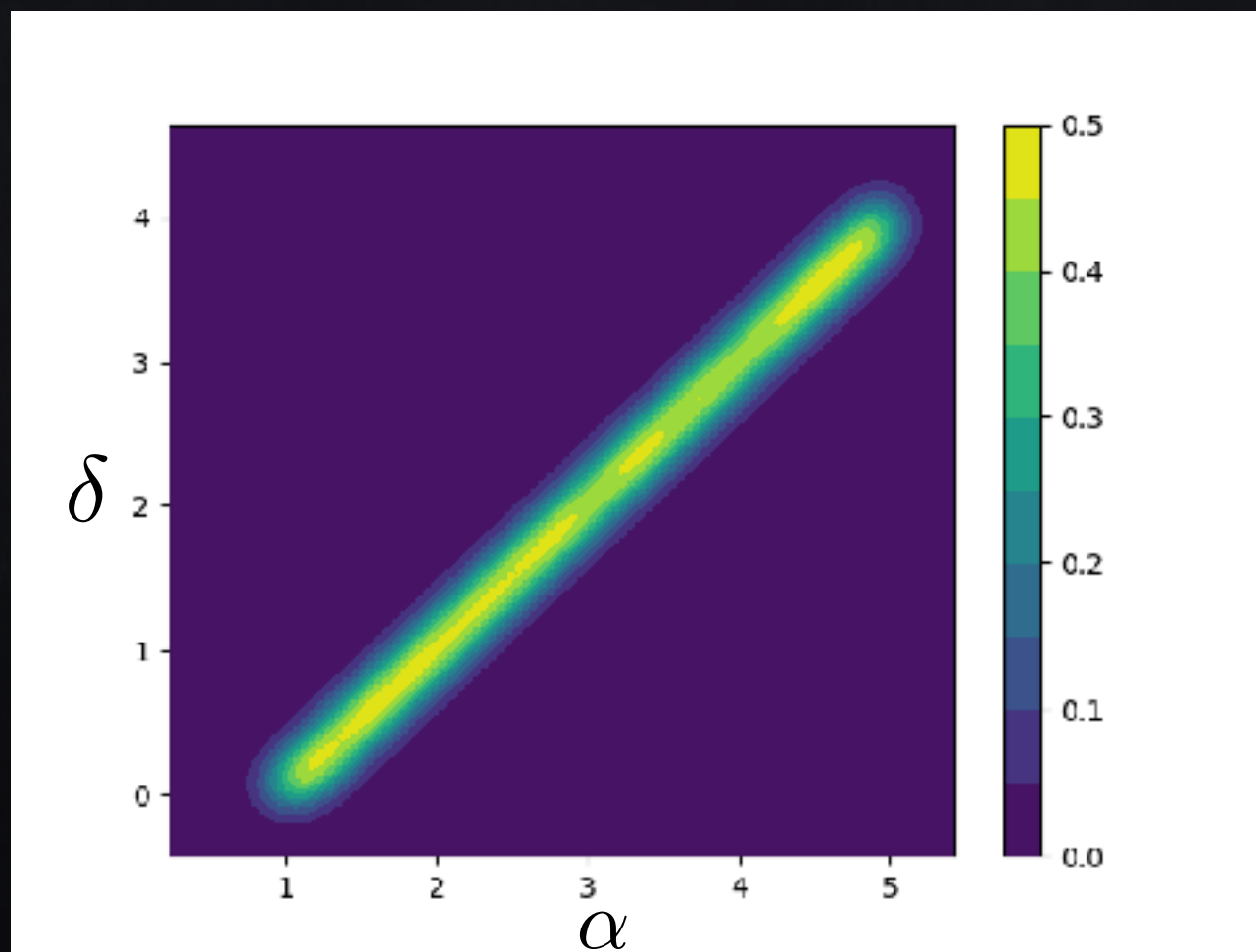
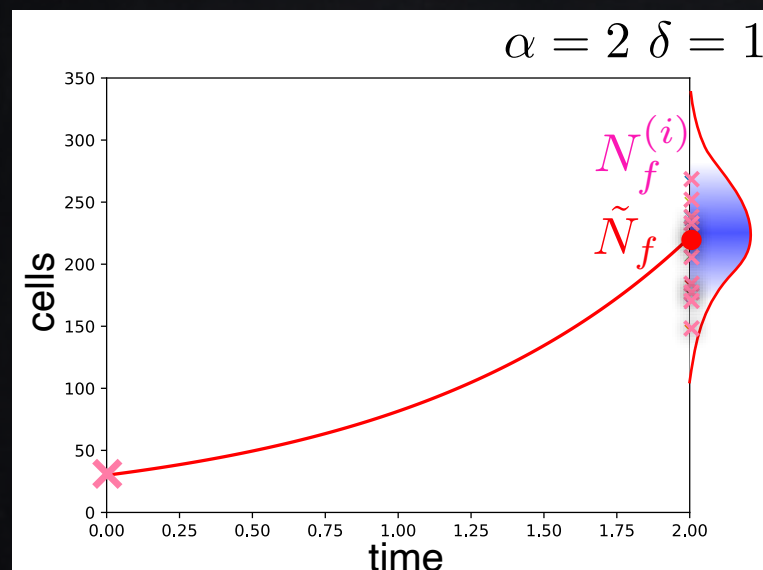


# Example 1.2: Population growth



$$\frac{dN(t)}{dt} = \alpha N - \delta N$$

## Posterior distributions

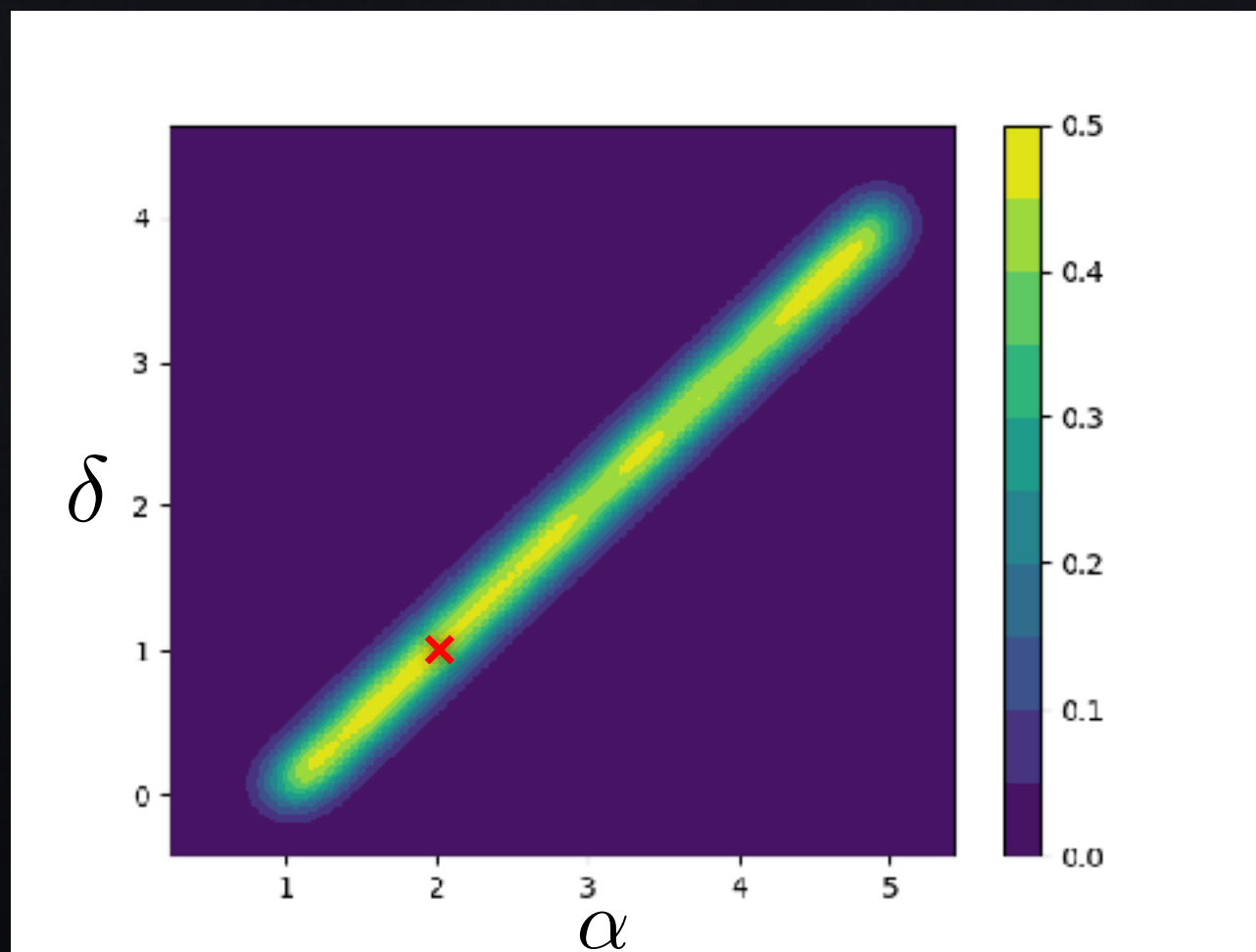
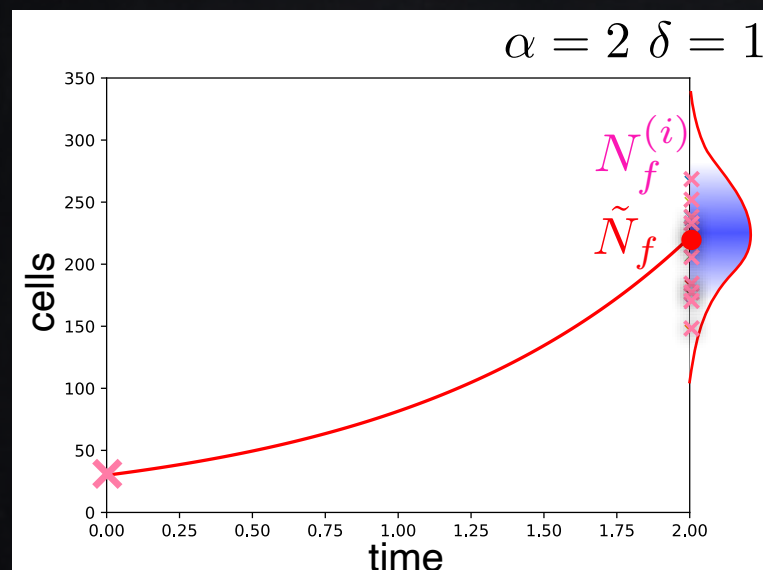


# Example 1.2: Population growth



$$\frac{dN(t)}{dt} = \alpha N - \delta N = (\alpha - \delta)N$$

## Posterior distributions

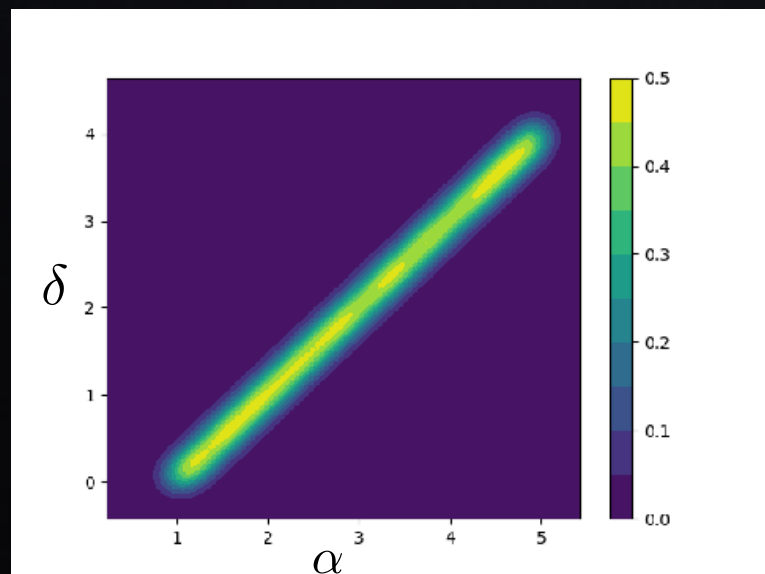
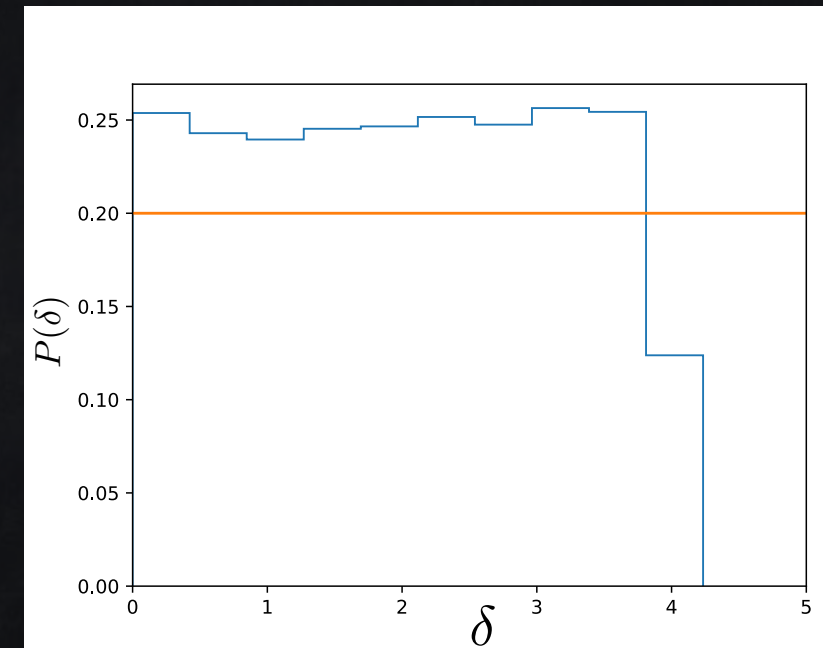
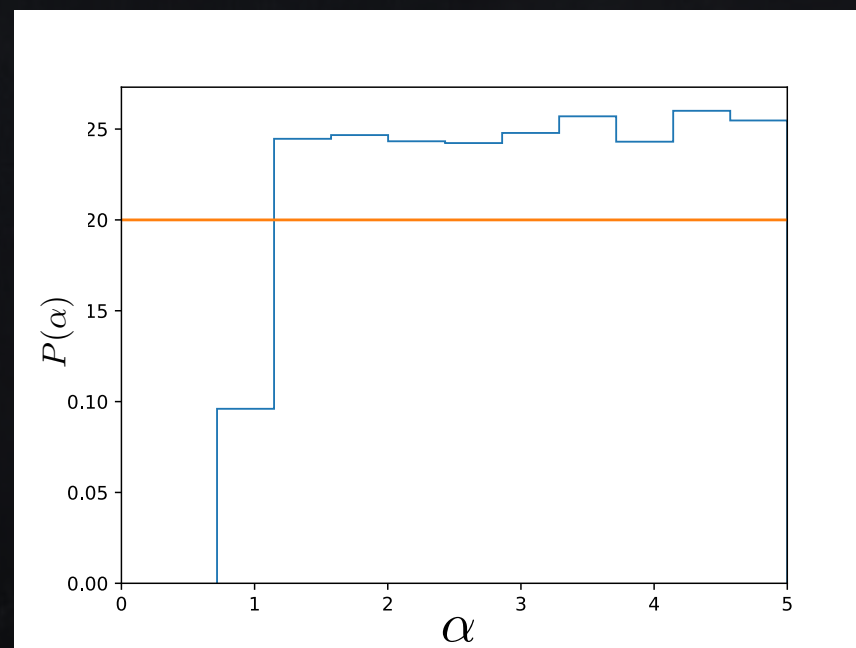
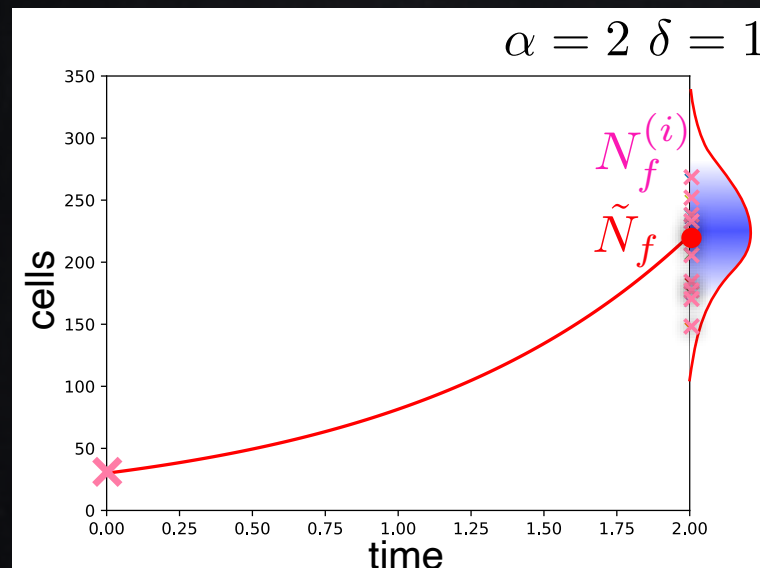


# Example 1.2: Population growth



$$\frac{dN(t)}{dt} = \alpha N - \delta N = (\alpha - \delta)N$$

## Posterior distributions



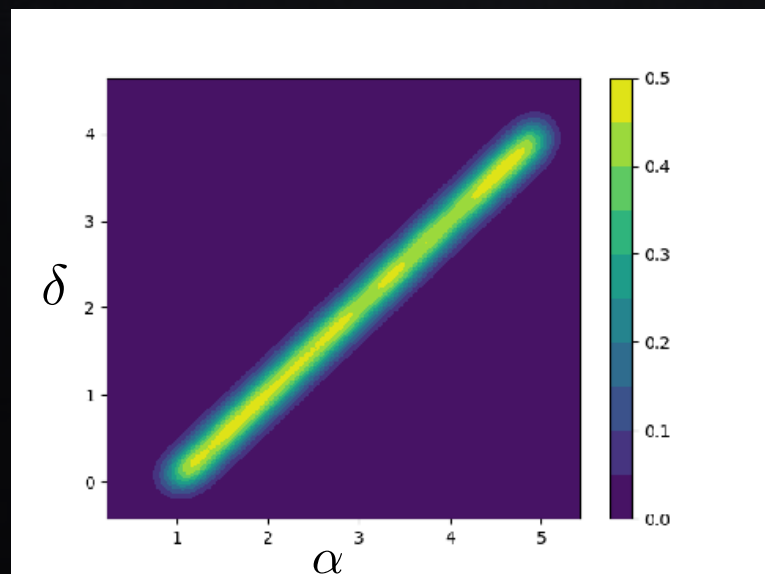
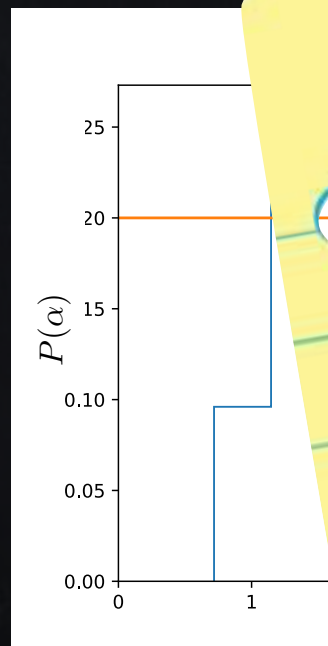
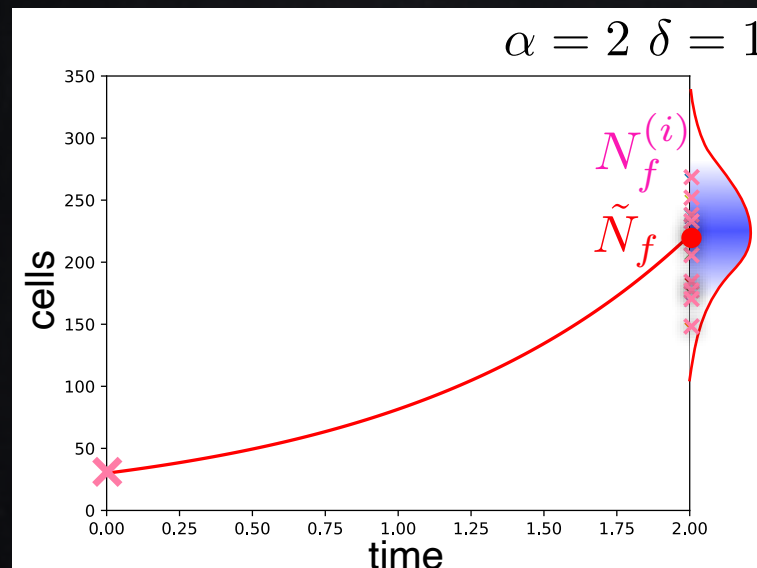
Bayesian inference yields how much information can we extract from the data with a model

# Example 1.2: Population growth



$$\frac{dN(t)}{dt} = \alpha N - \delta N = (\alpha - \delta)N$$

Posterior distributions



Bayesian inference  
we extract from

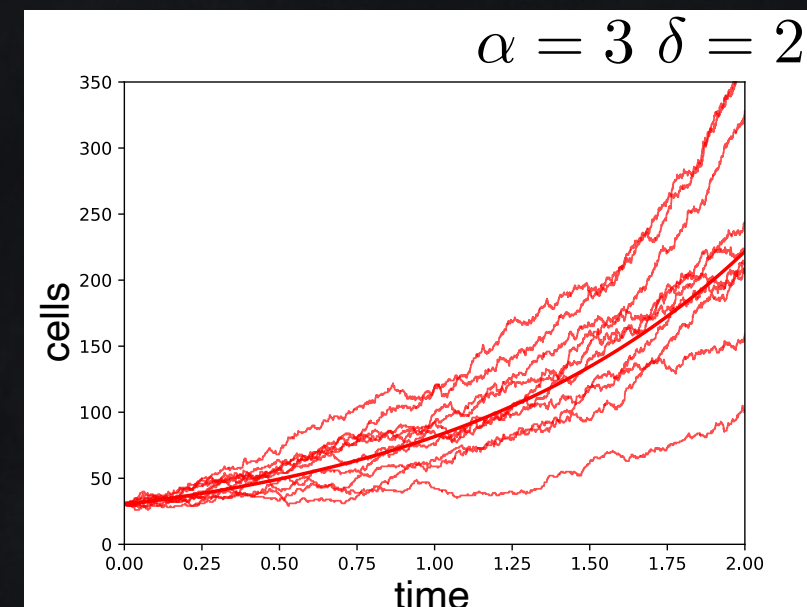
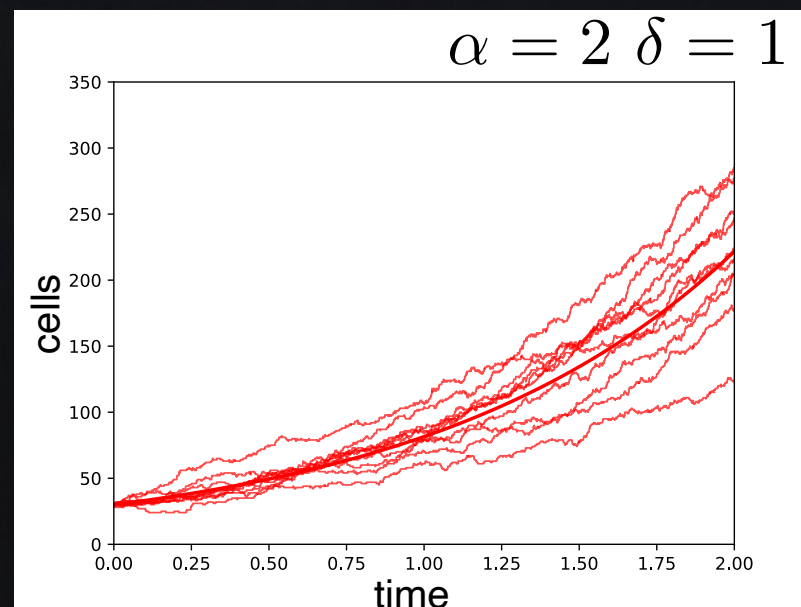
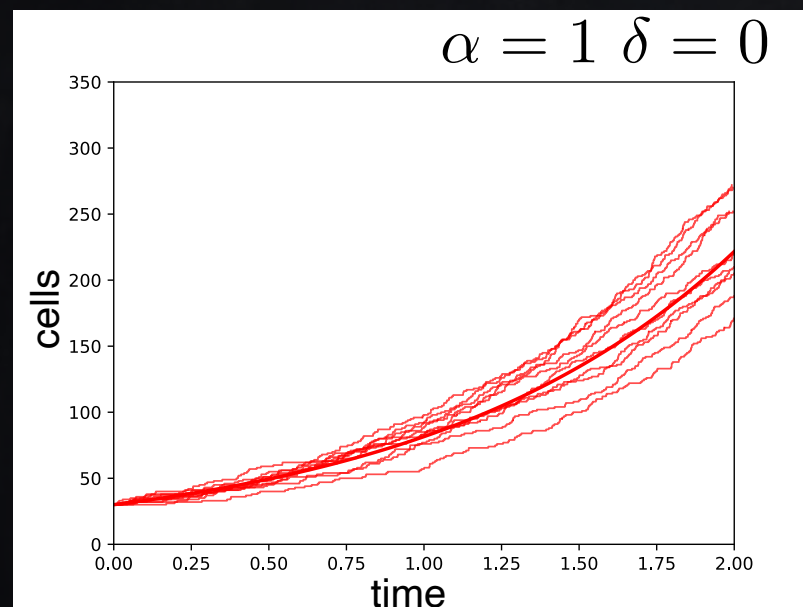
Question:

- Would this result change if we had a Gaussian prior distribution?
- What experiment would you propose to obtain narrower posteriors of the rates?

# Example 1.2: Population growth



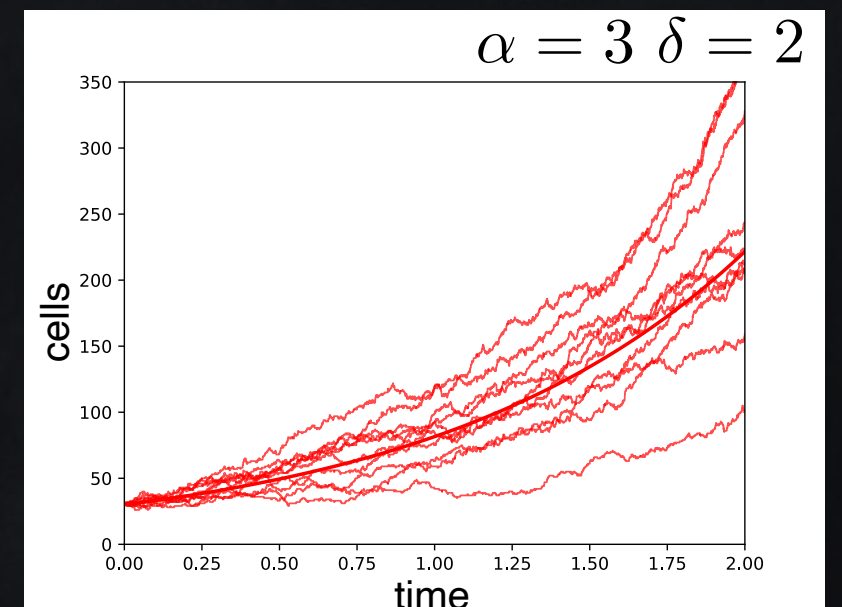
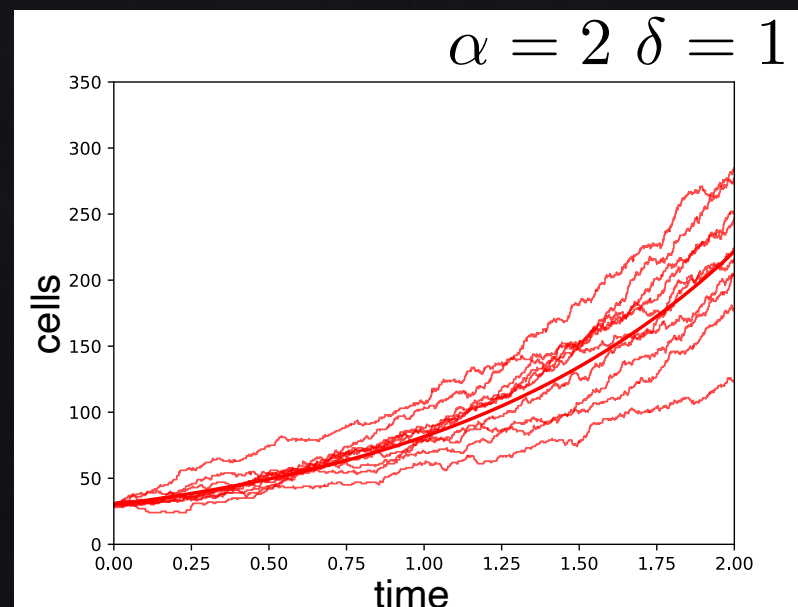
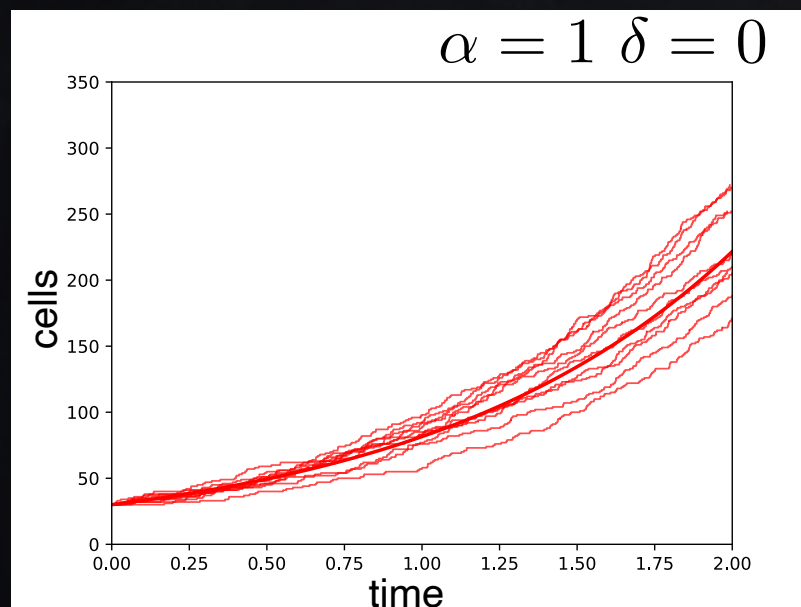
$$\frac{dN(t)}{dt} = \alpha N - \delta N = (\alpha - \delta)N$$



# Example 1.2: Population growth



$$\frac{dN(t)}{dt} = \alpha N - \delta N = (\alpha - \delta)N$$



Stochastic theory says...

$$\sigma(t) = N_0 \frac{\alpha + \delta}{\alpha - \delta} e^{2(\alpha - \delta)t} \left( 1 - e^{(\alpha - \delta)t} \right)$$

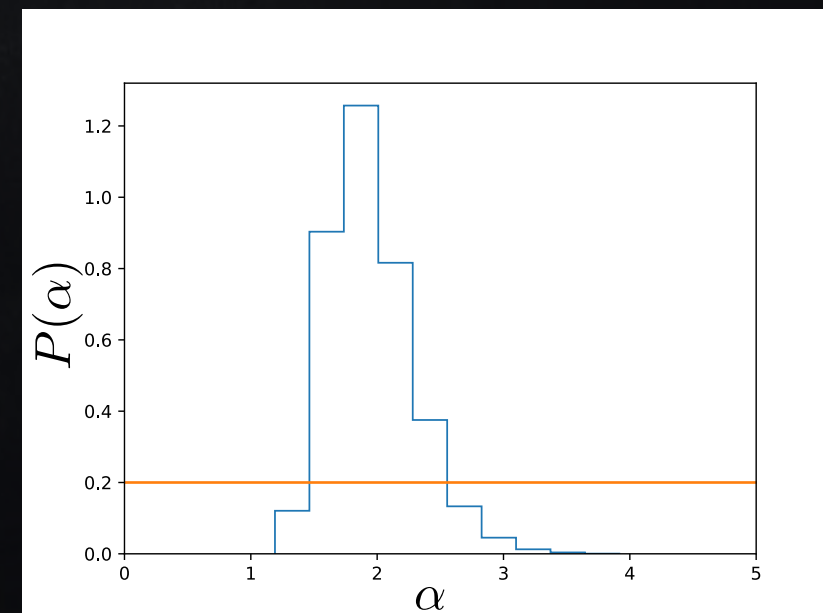
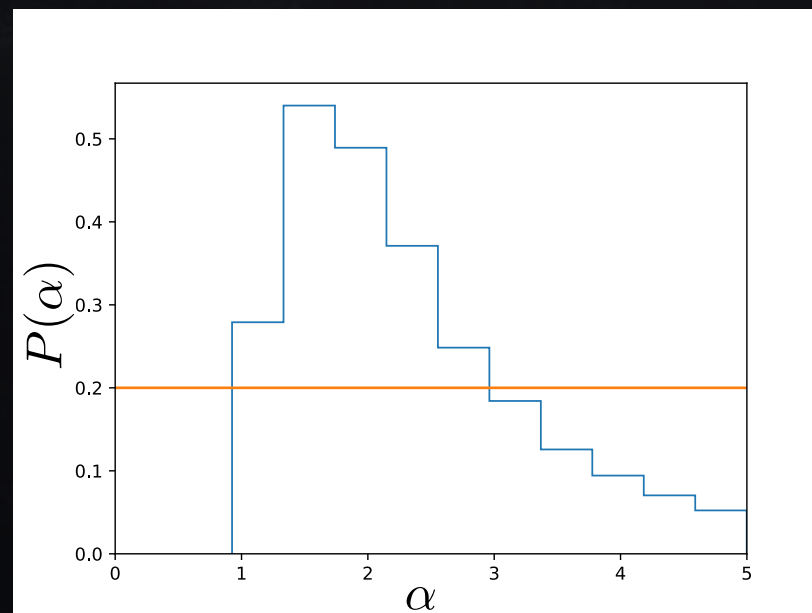
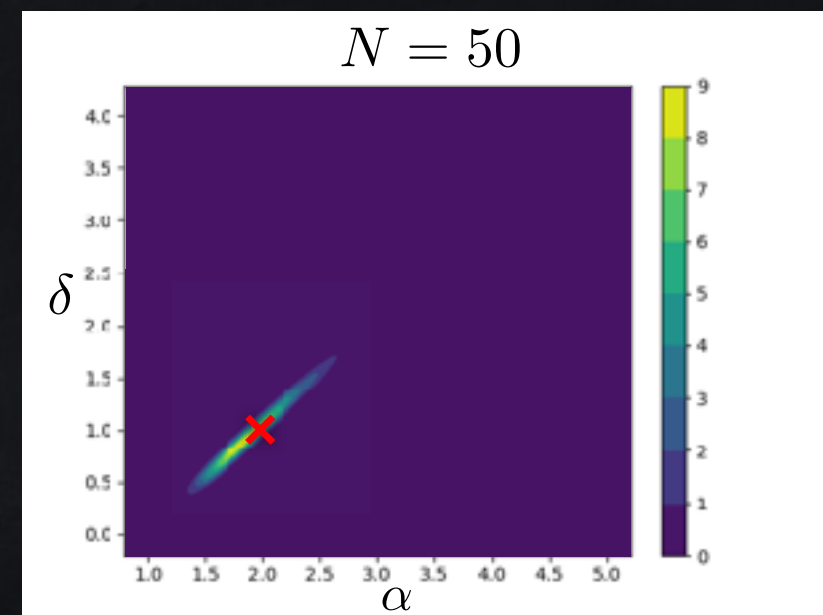
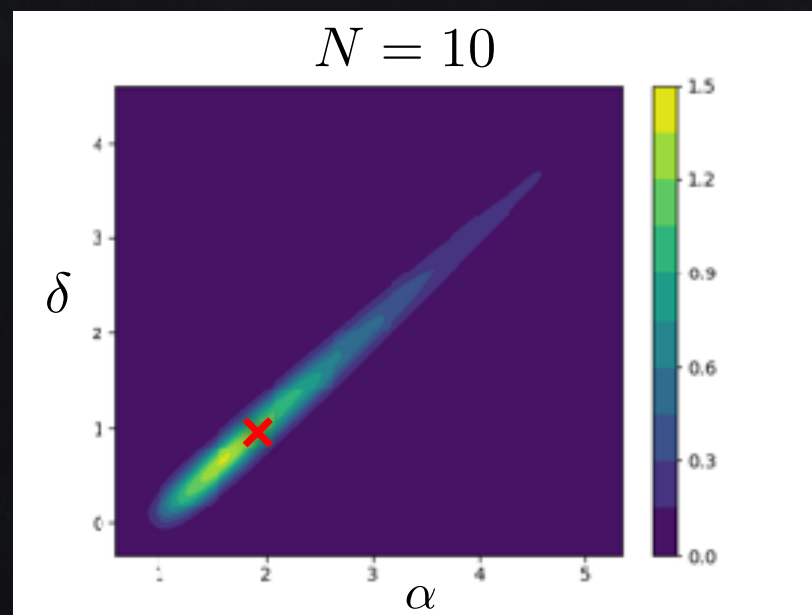
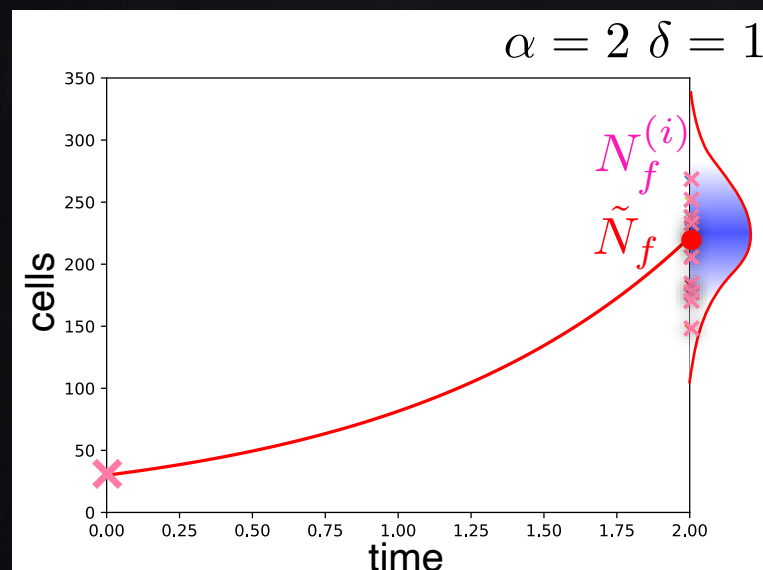
Likelihood

$$P(\text{data}|\alpha, \delta) = \prod_{i=1}^N P(\text{datapoint } i|\alpha, \delta) = \prod_{i=1}^N \mathcal{N}(\tilde{N}_f - \tilde{N}_f^{(i)}, \sigma)$$

# Example 1.2: Population growth



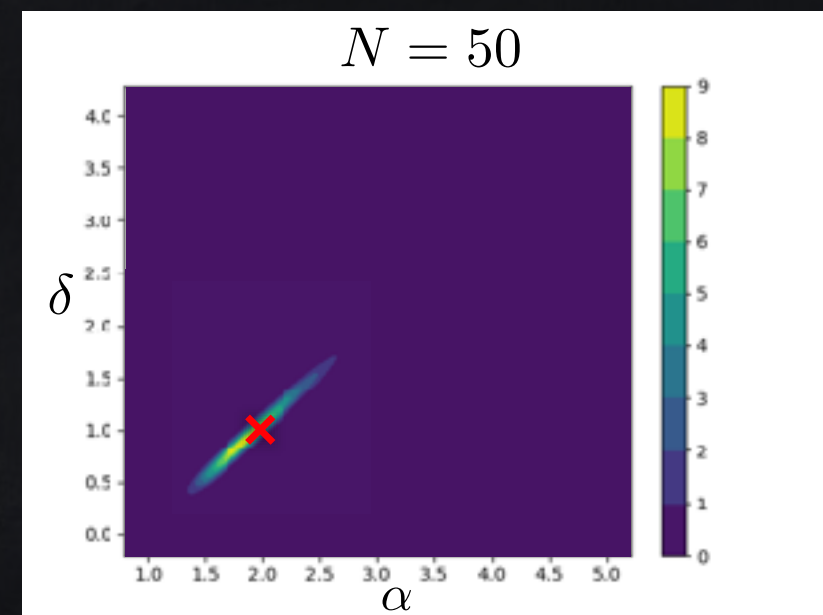
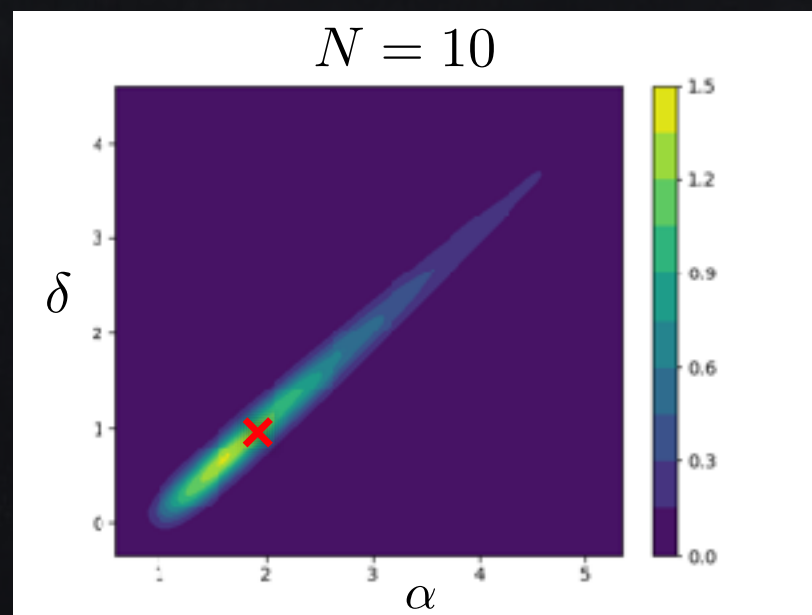
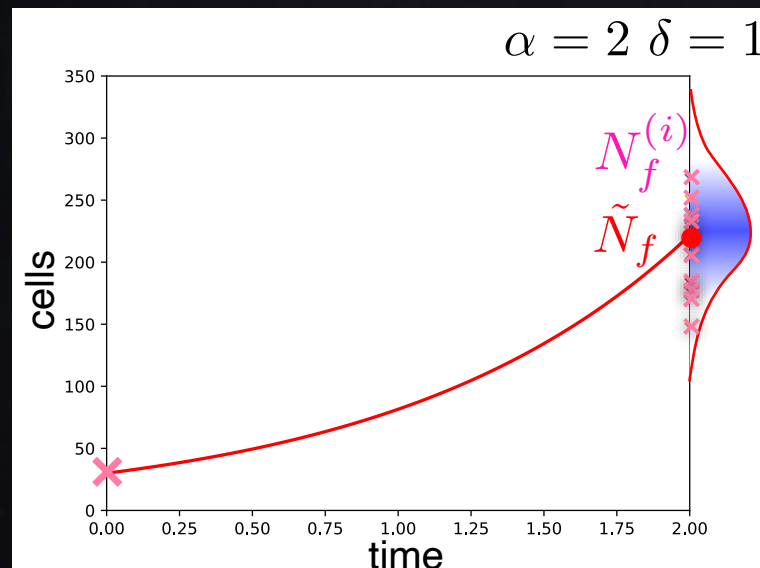
$$\frac{dN(t)}{dt} = \alpha N - \delta N = (\alpha - \delta)N$$



# Example 1.2: Population growth



$$\frac{dN(t)}{dt} = \alpha N - \delta N = (\alpha - \delta)N$$



Bayesian inference returns how much information can we extract from the data with a model

Therefore it can help us to...

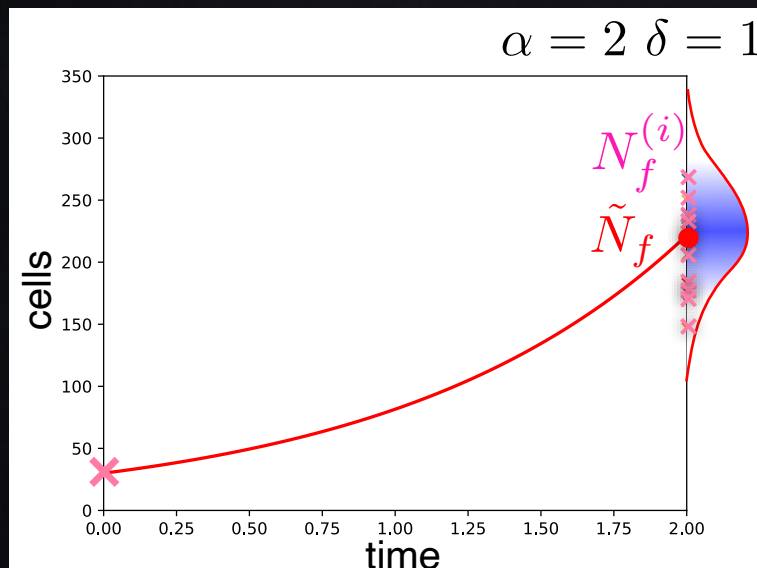


Design appropriate models e.g. include stochastic effects  
Design better experiments e.g. number of replicates

# Example 1.2: Population growth



$$\frac{dN(t)}{dt} = \alpha N - \delta N = (\alpha - \delta)N$$



Bayesian inference returns how much information can we extract from the data with a model

APPROXIMATE BAYESIAN COMPUTATION

Likelihood

$$P(\text{data}|\alpha, \delta) = \prod_{i=1}^N P(\text{datapoint } i|\alpha, \delta) = \prod_{i=1}^N \mathcal{N}(\tilde{N}_f - \tilde{N}_f^{(i)}, \sigma)$$

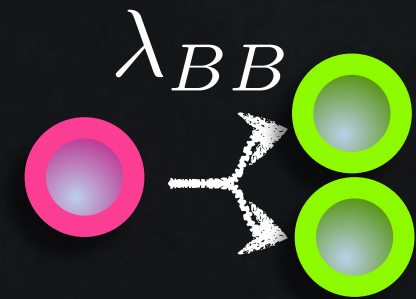
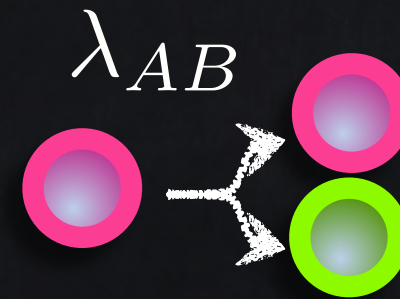
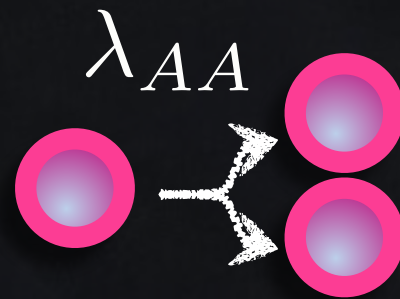
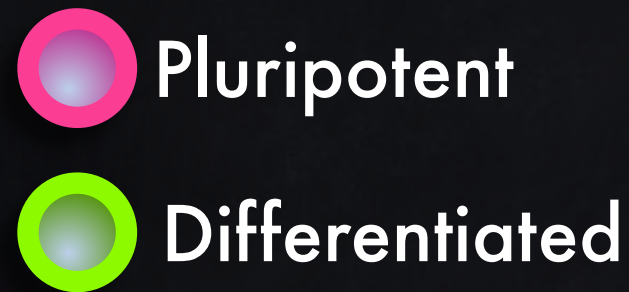
approximate  
(Likelihood - free)

$$= \prod_{i=1}^N \sum_{j=0}^{\min(N_f^{(i)}, N_0)} \binom{N_0}{j} \binom{N_0 + N_f^{(i)} - j - 1}{N_0 - 1} \alpha^{N_0 - j} \beta^{N_f^{(i)} - j} (1 - \alpha - \beta)^j$$

exact

Beyond

# Example 1: Population growth



Physical Biology

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Coupled differentiation and division of embryonic stem cells inferred from clonal snapshots

To cite this article before publication: Liam J Ruske *et al* 2020 *Phys. Biol.* in press <https://doi.org/10.1088/1478-3975/aba041>

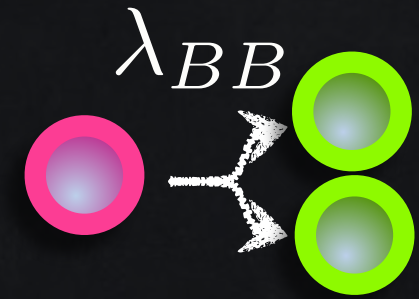
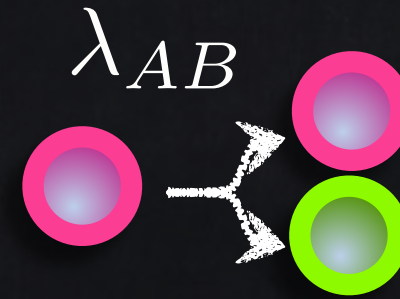
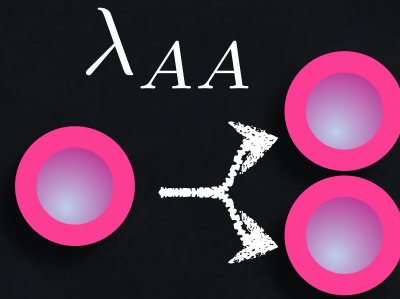
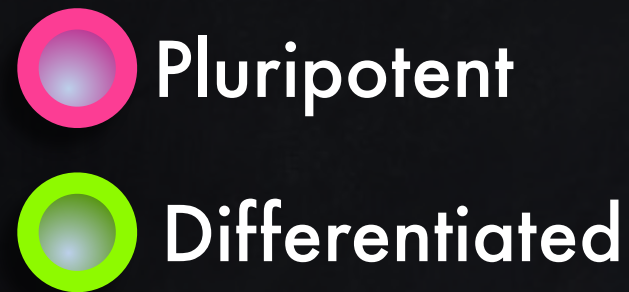
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Beyond

# Example 1: Population growth



Can we distinguish differentiation mechanisms?

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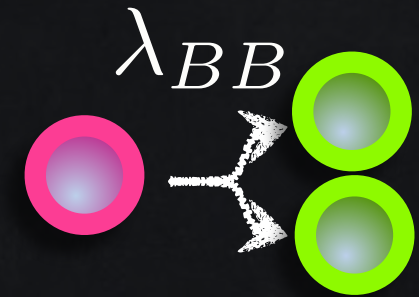
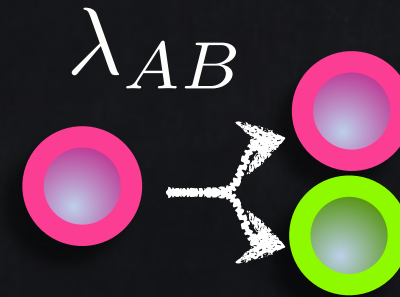
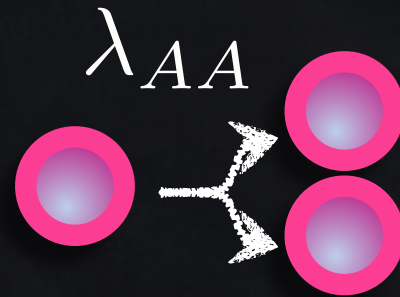
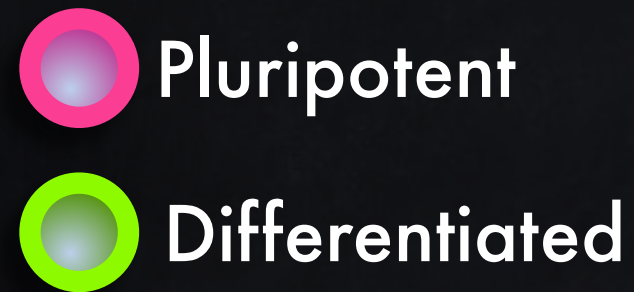
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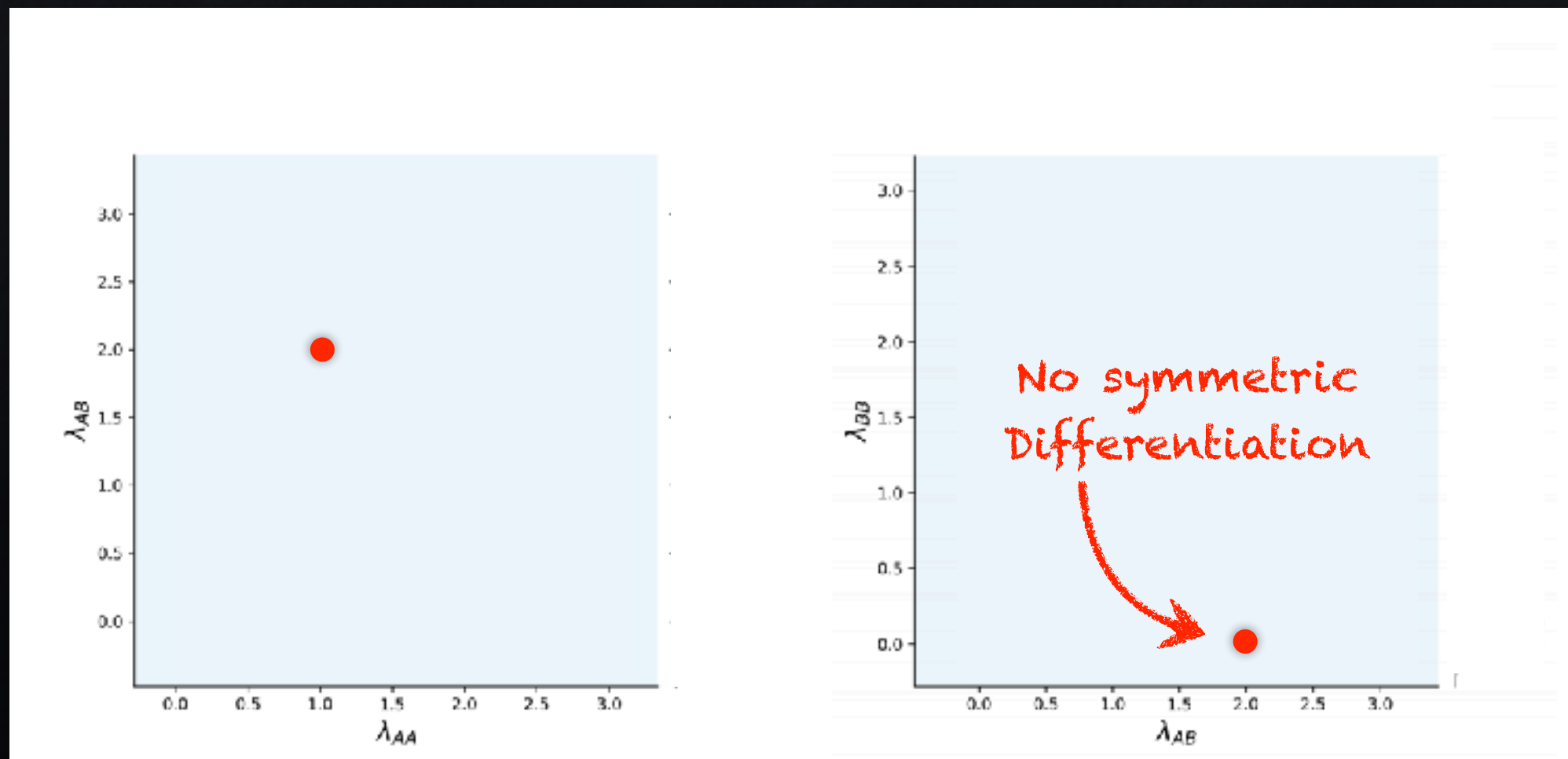
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# Beyond

## Example 1: Population growth



Can we distinguish differentiation mechanisms?

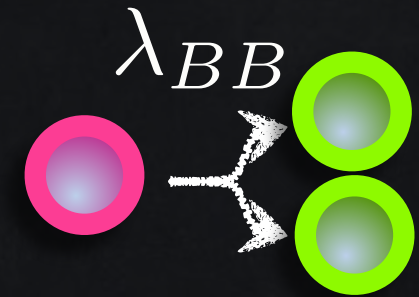
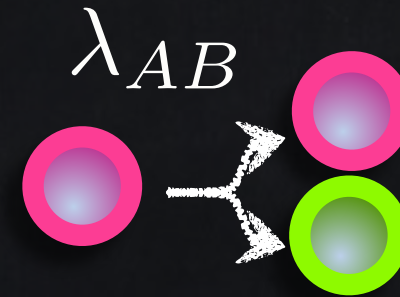
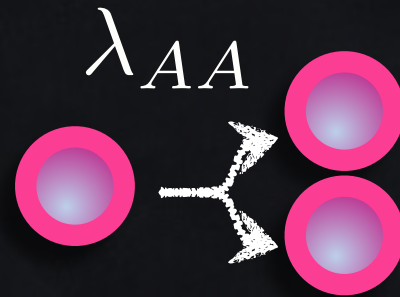


Beyond

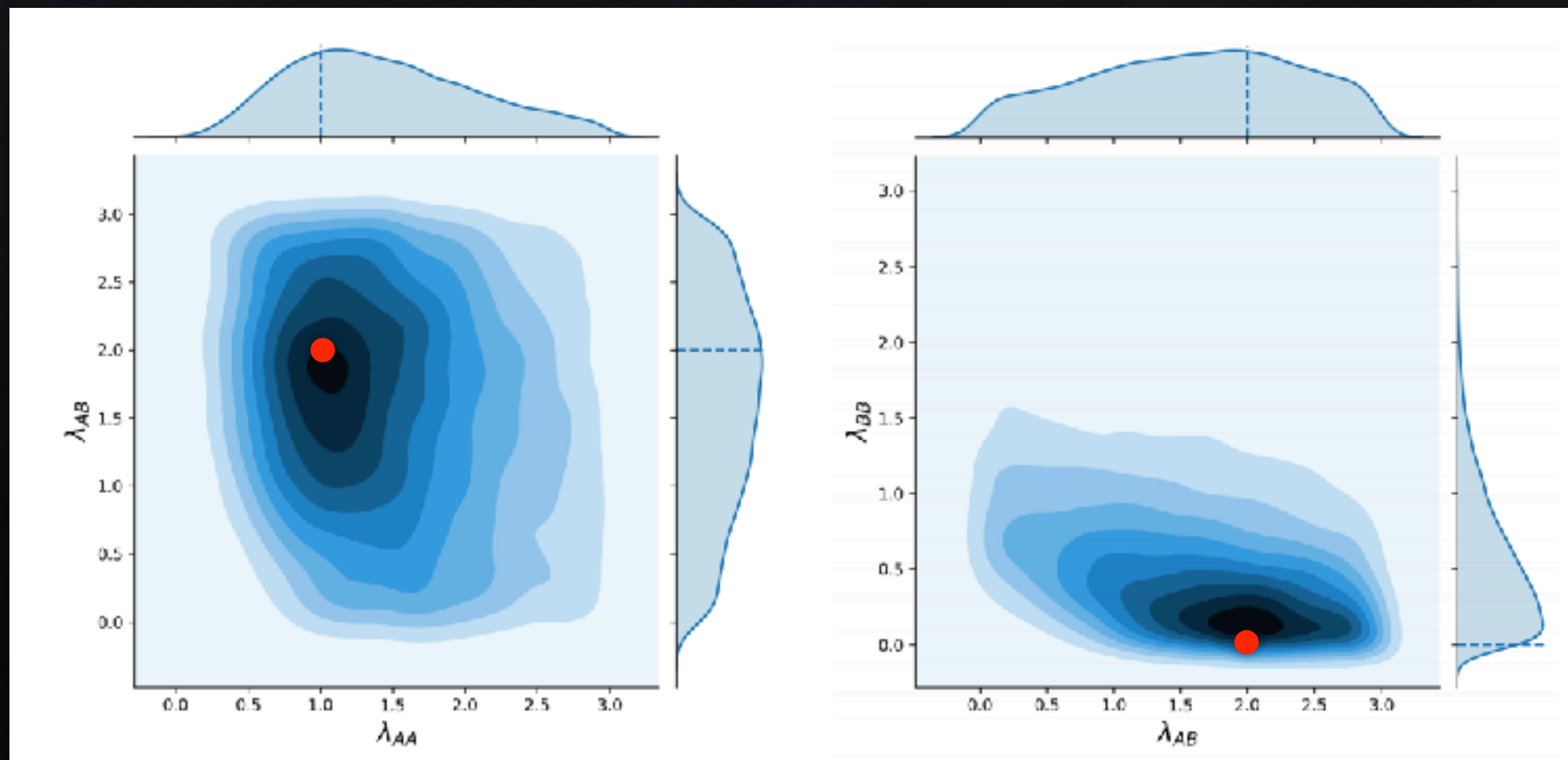
# Example 1: Population growth

 Pluripotent

 Differentiated

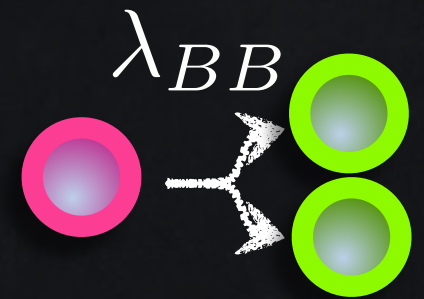
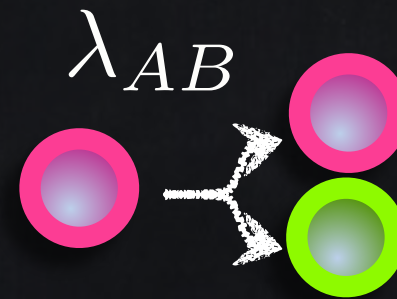
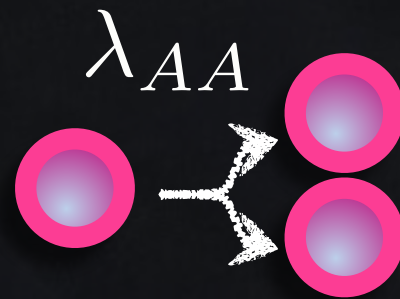
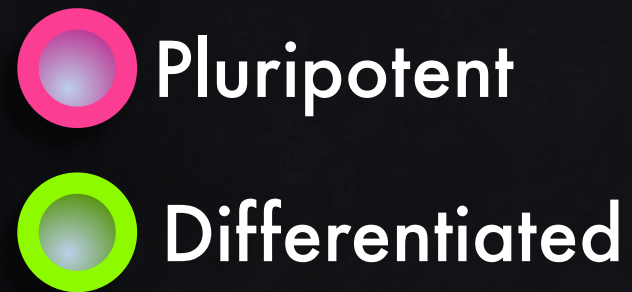


Can we distinguish differentiation mechanisms?



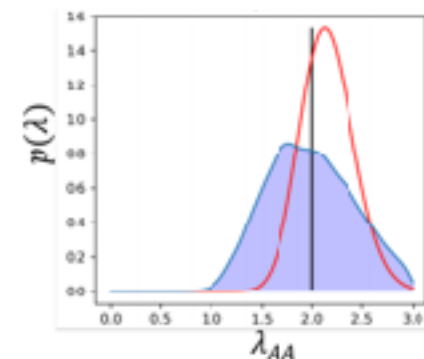
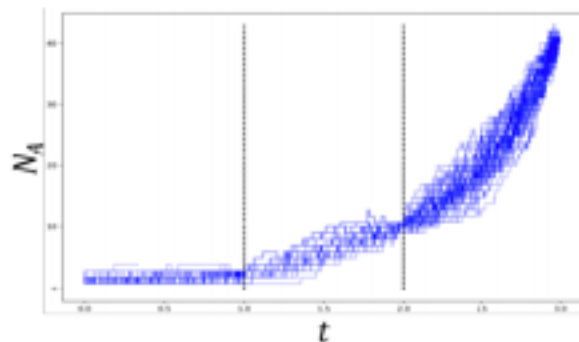
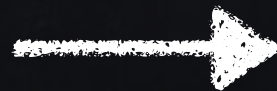
# Beyond

## Example 1: Population growth



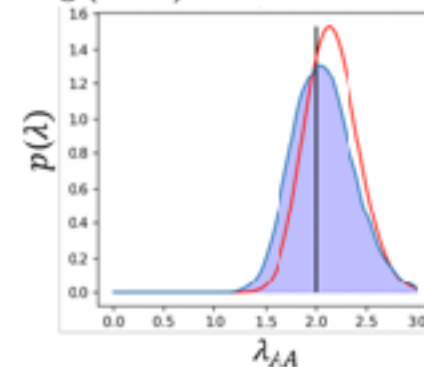
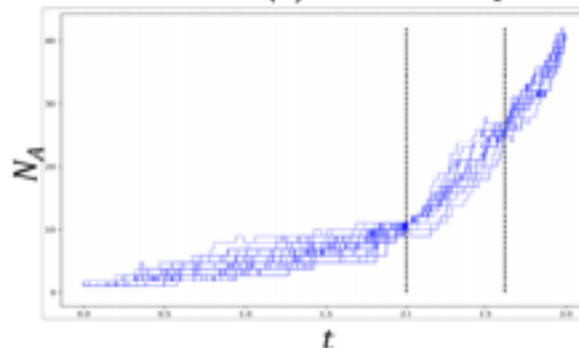
Can we distinguish differentiation mechanisms?  
Is there an optimal set of sampling times?

Uniform  
sampling



(a) Uniform snapshot spacing ( $z = 0$ )

Optimal  
sampling



# Bayesian inference methods for ecology

## Theory

What is Bayesian inference?

Probability 101

Bayes' theorem in biology

Priors and likelihood

Interpretation of posteriors

Markov Chain Monte Carlo

## Applications

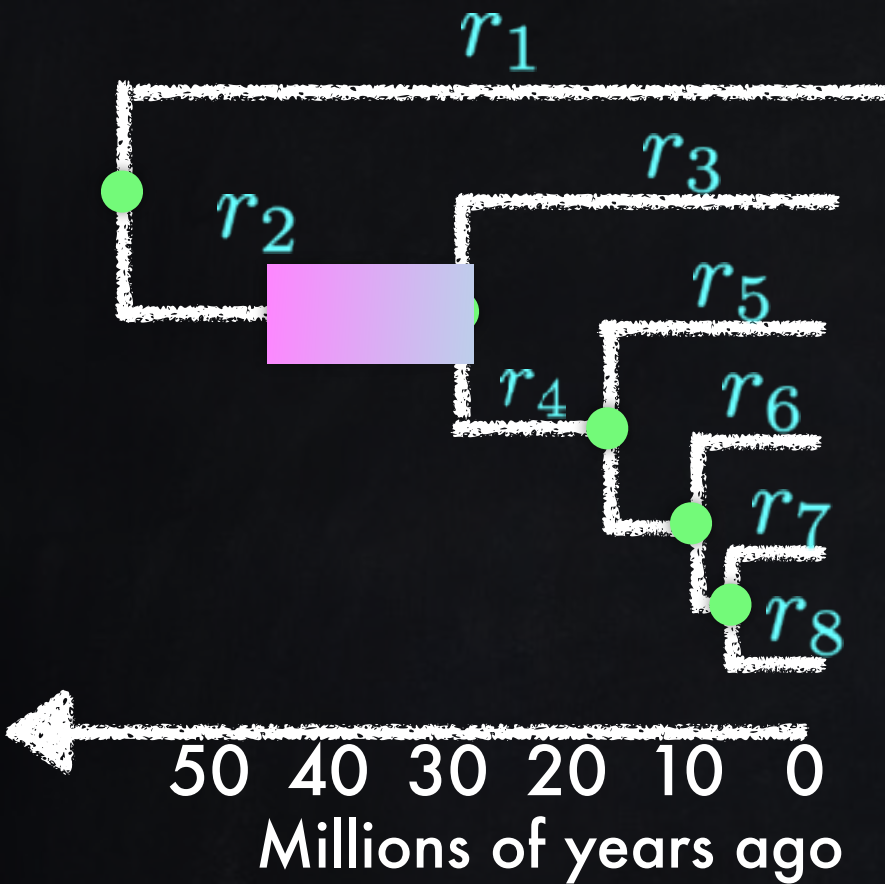
(to cell/developmental biology)

Proliferation and  
differentiation of a  
stem cell population

Phylogeny and model  
selection

Morphogen patterning  
of embryonic tissues

# Phylogeny



Marmoset  
Macaque  
Orangutan  
Gorilla  
Chimpanzee  
Human

Data

DNA alignment  
Fossils/Events

Uncertainty

Mutation variability  
Inaccurate bounds

Tree =  $\{t_i\}$

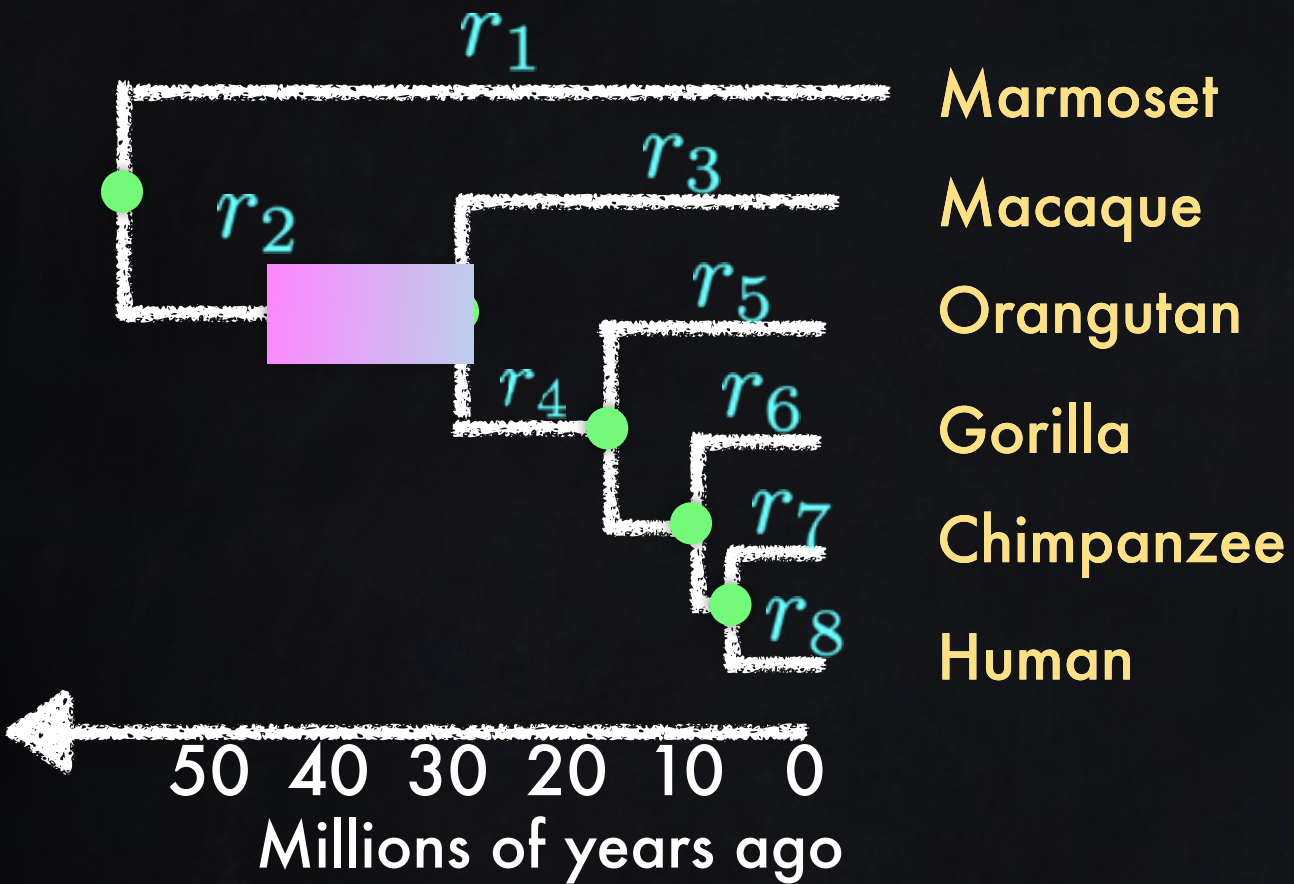
$P(t_i|data)$

Speciation times

Mutation rates  
model parameters

$P(t_i, r_i, \theta|data)$

# Phylogeny



Data

DNA alignment

Fossils/Events

Uncertainty

Mutation variability

Inaccurate bounds

Tree =  $\{t_i\}$

$P(t_i|data)$

Speciation times

$$P(t_i, r_i, \theta | data) \propto P(data | r_i, t_i, \theta) P(r_i, t_i, \theta)$$

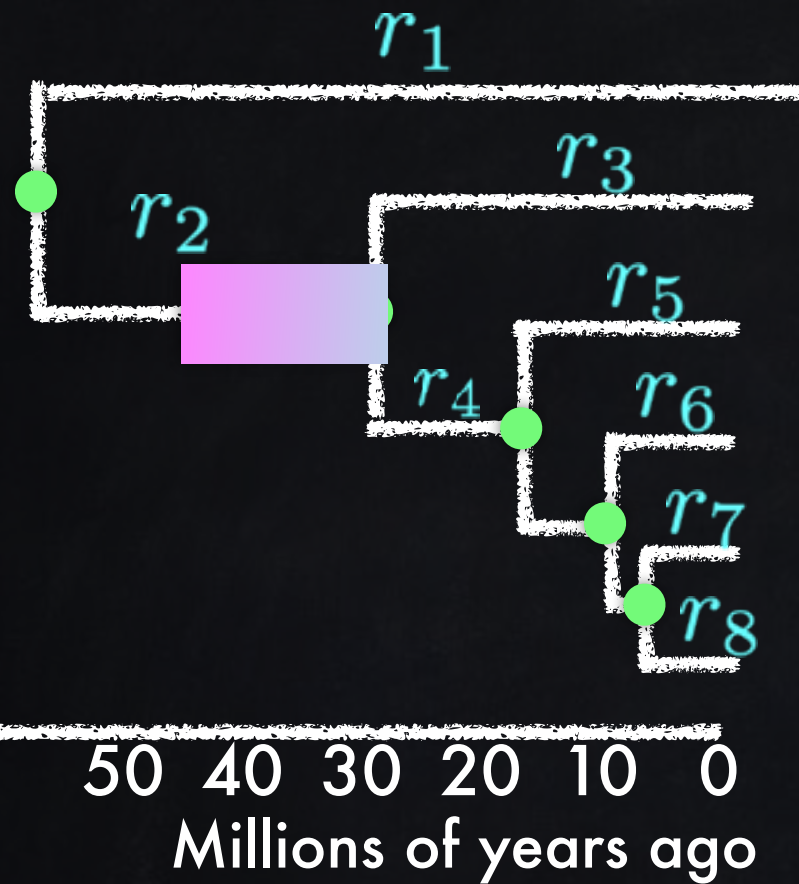
$$\propto P(data | r_i, t_i) P(r_i | t_i, \theta) P(t_i | \theta) P(\theta)$$

Nucleotide  
substitution  
model

Molecular clock  
model

Divergence time  
model

# Phylogeny



Marmoset  
Macaque  
Orangutan  
Gorilla  
Chimpanzee  
Human



Bayesian inference allows for nested models that capture the complexity of underlying structure/correlations:  
**Hierarchical models**

Tree =  $\{t_i\}$

$P(t_i|data)$

Speciation times

Mutation rates

model  
parameters

$$P(t_i, r_i, \theta | data) \propto P(data | r_i, t_i, \theta) P(r_i, t_i, \theta)$$

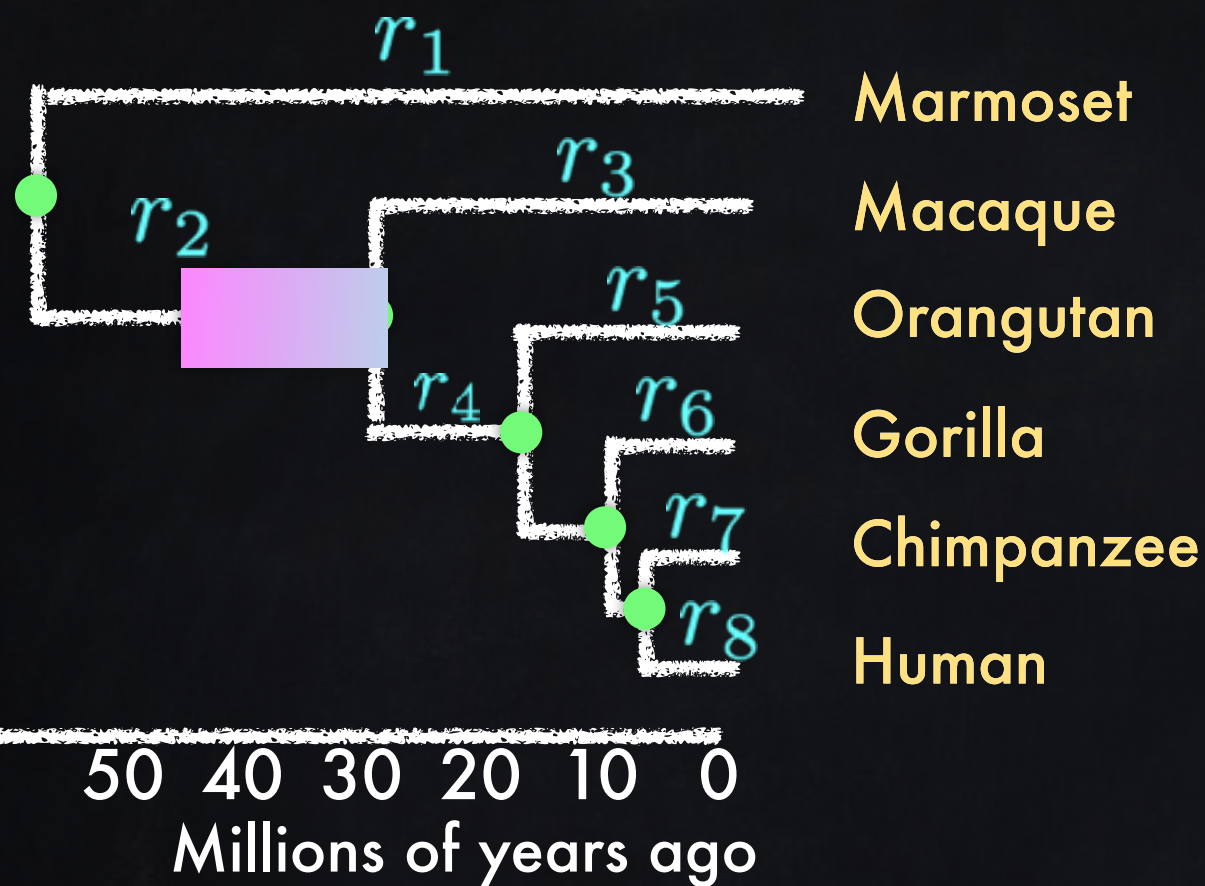
$$\propto P(data | r_i, t_i) P(r_i | t_i, \theta) P(t_i | \theta) P(\theta)$$

Nucleotide  
substitution  
model

Molecular clock  
model

Divergence time  
model

# Phylogeny



Bayesian inference allows for nested that capture the complexity of underlying structure/correlations:  
**Hierarchical models**

Mutation rates  $\uparrow$  model parameters  $\nearrow$

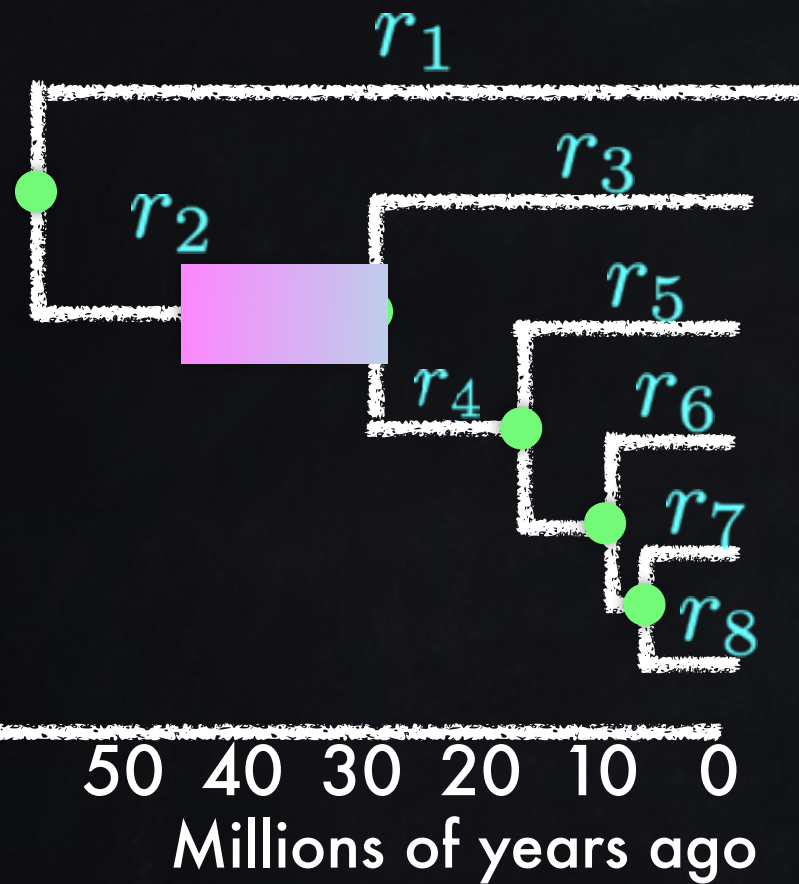
$$P(t_i, r_i, \theta | data) \propto P(data | r_i, t_i, \theta) P(r_i, t_i, \theta)$$

$$\propto P(data | r_i, t_i) P(r_i | t_i, \theta) P(t_i | \theta) P(\theta)$$

$$\propto P(data | r_i, t_i) P(r_i | t_i, \theta, \theta_M) P(t_i | \theta) P(\theta, \theta_M)$$

Model selection parameter  $\theta_M$

# Phylogeny



Marmoset  
Macaque  
Orangutan  
Gorilla  
Chimpanzee  
Human



Bayesian inference allows for nested that capture the complexity of underlying structure/correlations:  
**Hierarchical models**



$$\frac{P(\theta_1|data)}{P(\theta_2|data)} = \text{Bayes' factor}$$

Credibility among models

Mutation rates  $\uparrow$   
model parameters  $\nearrow$

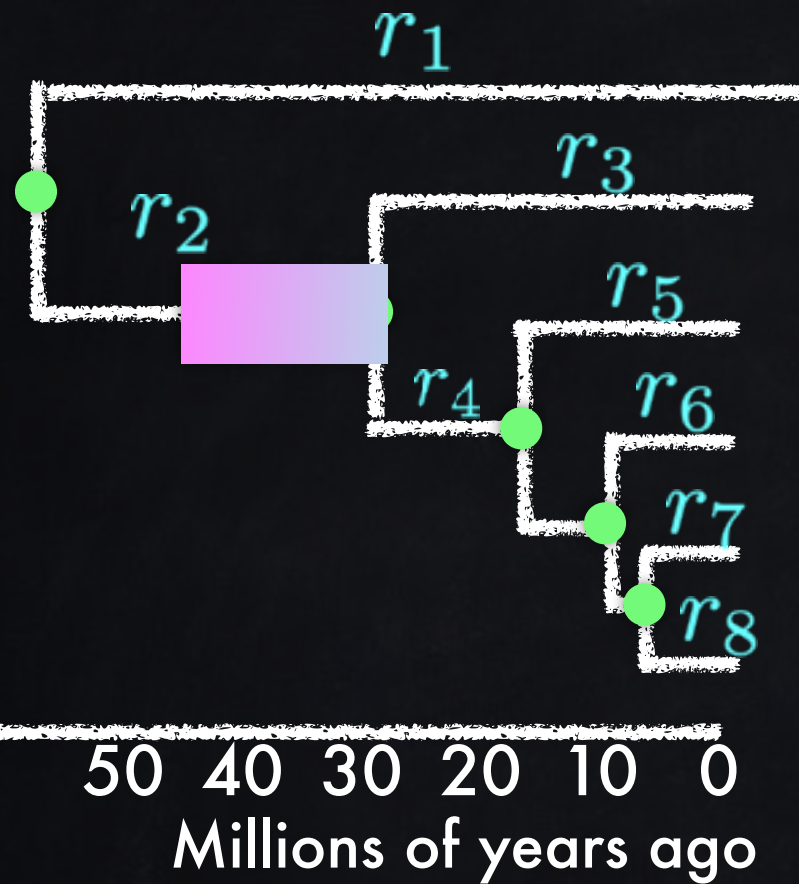
$$P(t_i, r_i, \theta|data) \propto P(data|r_i, t_i, \theta)P(r_i, t_i, \theta)$$

$$\propto P(data|r_i, t_i)P(r_i|t_i, \theta)P(t_i|\theta)P(\theta)$$

$$\propto P(data|r_i, t_i)P(r_i|t_i, \theta, \theta_M)P(t_i|\theta)P(\theta, \theta_M)$$

Model selection parameter  $\nwarrow$

# Phylogeny



Marmoset  
Macaque  
Orangutan  
Gorilla  
Chimpanzee  
Human



Bayesian inference allows for nested that capture the complexity of underlying structure/correlations:  
**Hierarchical models**



Bayesian inference allows model selection



$$\frac{P(\theta_1|data)}{P(\theta_2|data)} = \text{Bayes' factor}$$

Credibility among models

Mutation rates  $\uparrow$   
model parameters  $\nearrow$

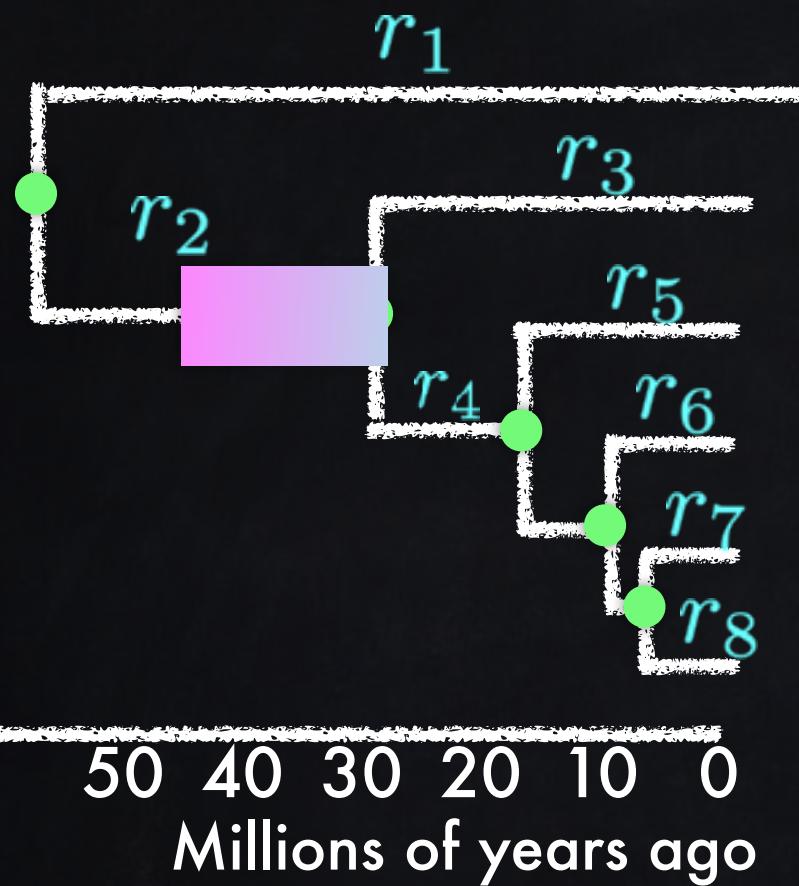
$$P(t_i, r_i, \theta|data) \propto P(data|r_i, t_i, \theta)P(r_i, t_i, \theta)$$

$$\propto P(data|r_i, t_i)P(r_i|t_i, \theta)P(t_i|\theta)P(\theta)$$

$$\propto P(data|r_i, t_i)P(r_i|t_i, \theta, \theta_M)P(t_i|\theta)P(\theta, \theta_M)$$

Model selection parameter

# Phylogeny



Marmoset  
Macaque  
Orangutan  
Gorilla  
Chimpanzee  
Human



Bayesian inference allows for nested that capture the complexity of underlying structure/correlations:  
**Hierarchical models**



Bayes  
mode



$$\frac{P(\theta_1|data)}{P(\theta_2|data)}$$

Cre

Mutation rates  
model parameters

$$P(t_i, r_i, \theta|data) \propto P(data|r_i, t_i, \theta)P(r_i, t_i)$$

$$\propto P(data|r_i, t_i)P(r_i|t_i, \theta)$$

$$\propto P(data|r_i, t_i)P(r_i|t_i, \theta,$$

Model sele



## The Molecular Evolutionary Clock

Theory and Practice

# Example 3: Developmental GRNs / Neural tube patterning



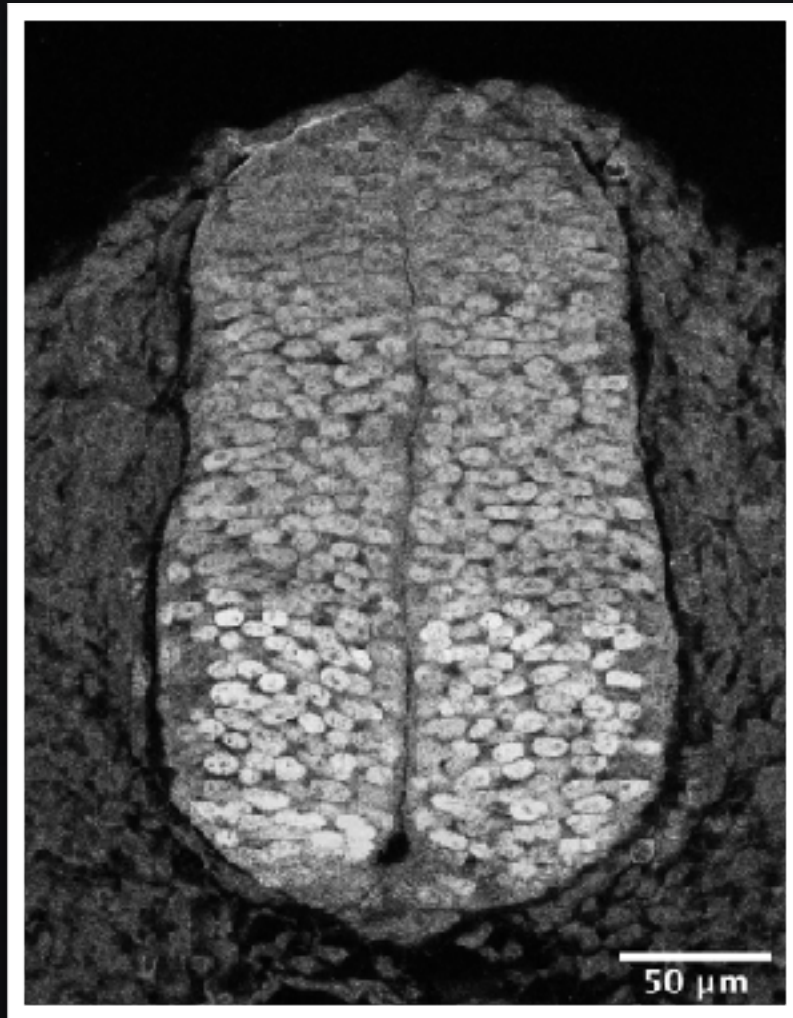
Which?

How?

Where?

When?

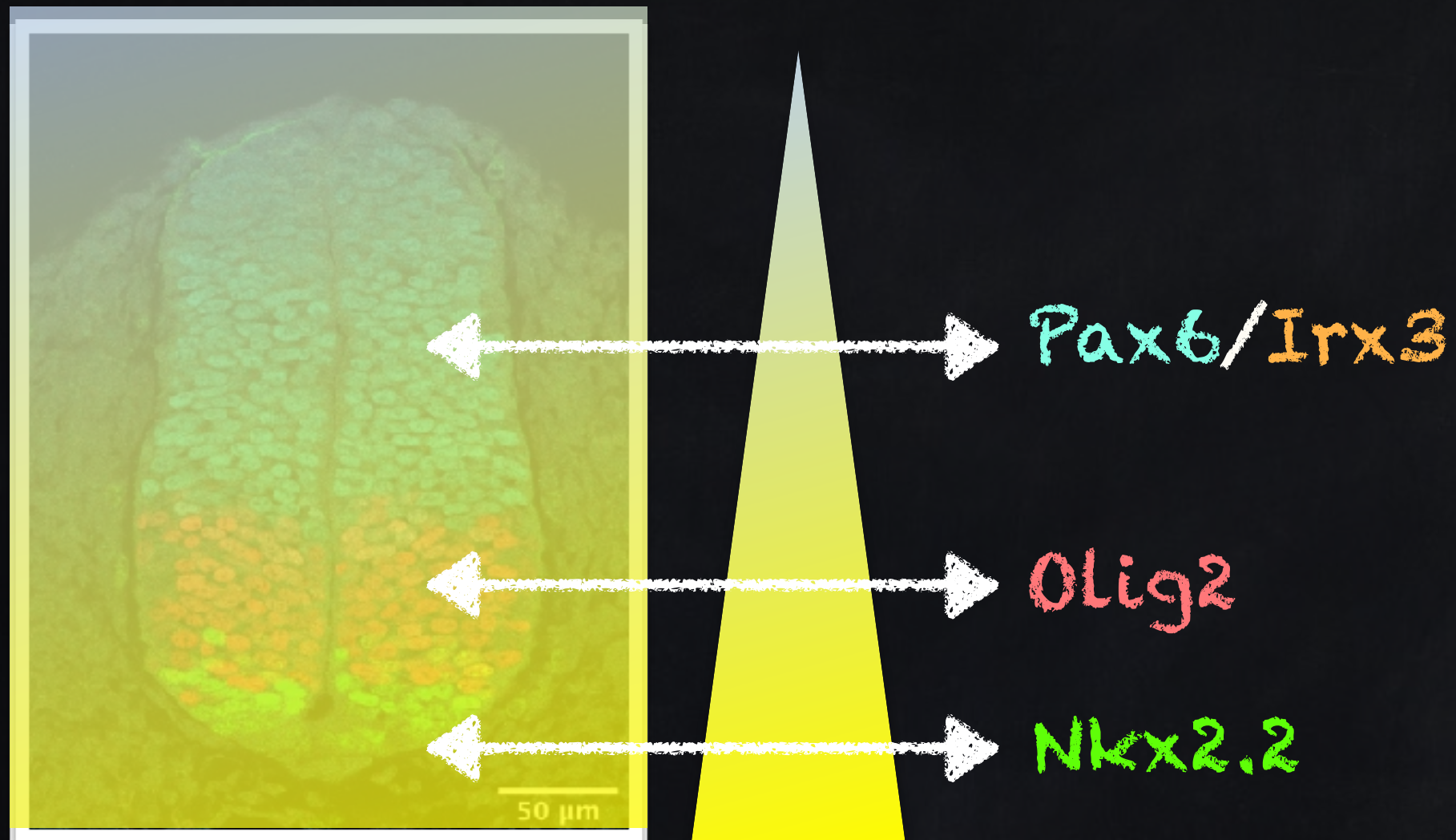
# Example 3: Developmental GRNs / Neural tube patterning



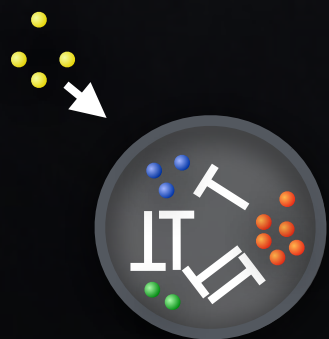
Neural tube



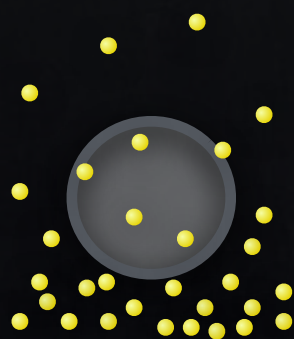
# Example 3: Developmental GRNs / Neural tube patterning



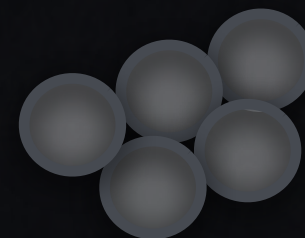
Sonic  
Hedgehog



Single cell  
regulation

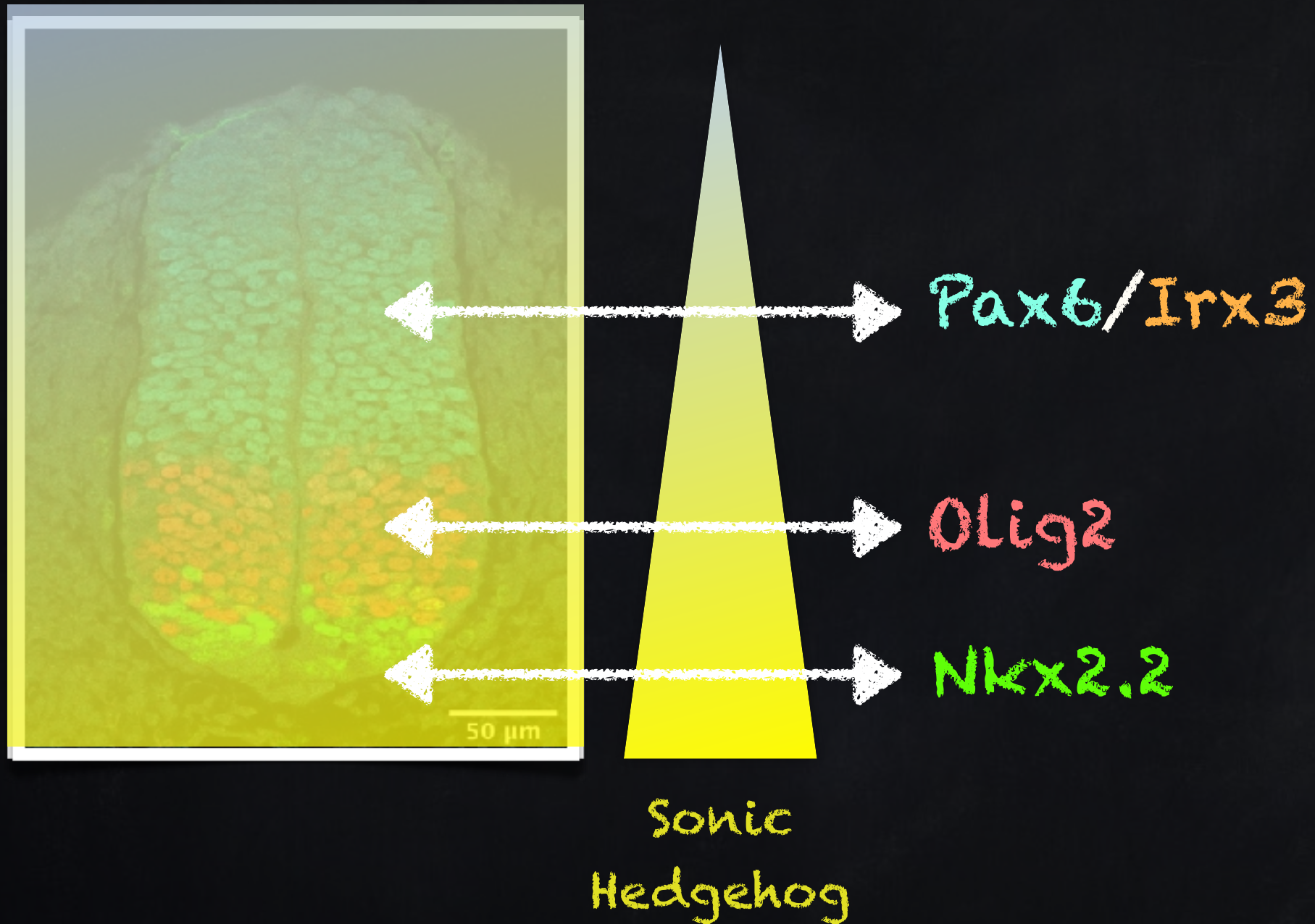


Signal  
diffusion



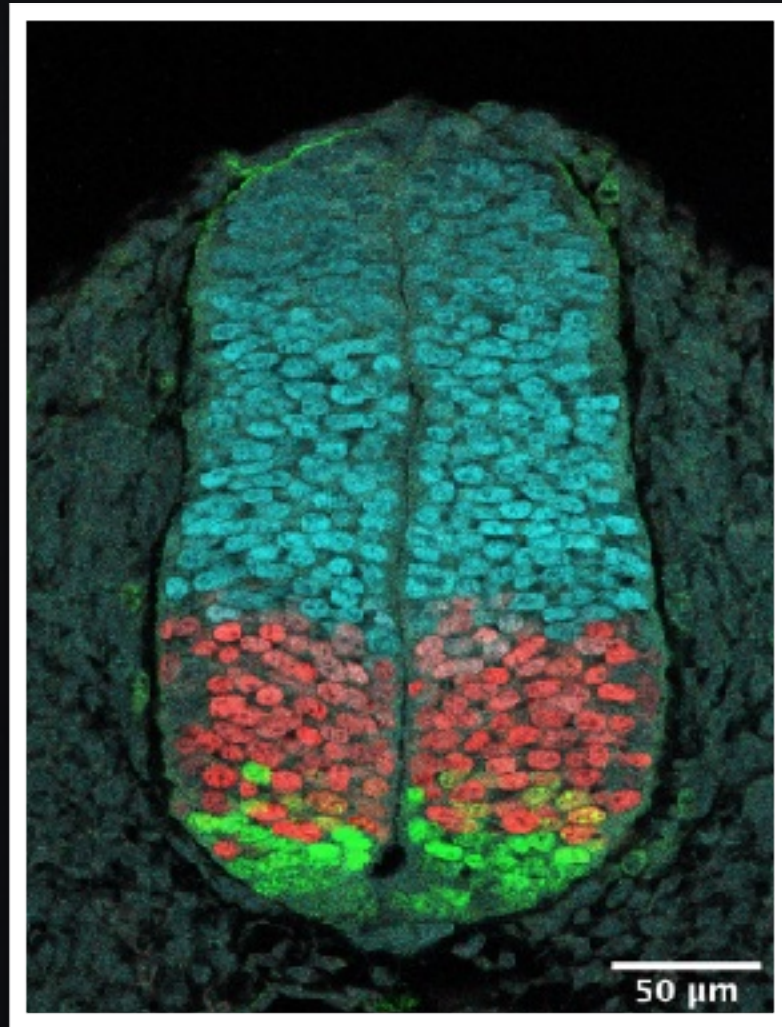
Tissue  
dynamics

# Example 3: Developmental GRNs / Neural tube patterning



How does gene regulation  
control the patterning mechanism?

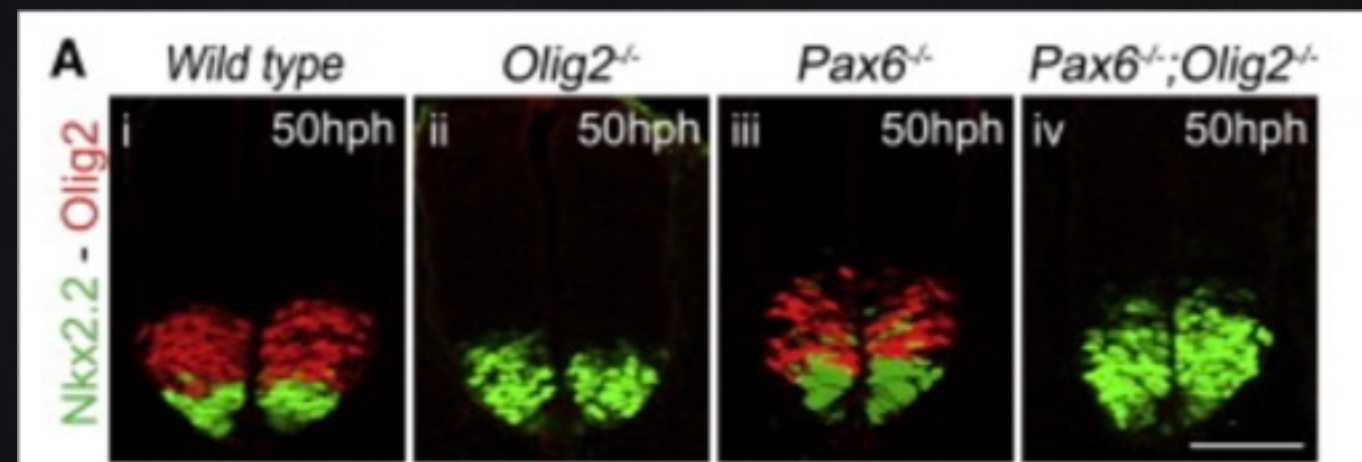
# Example 3: Developmental GRNs / Neural tube patterning



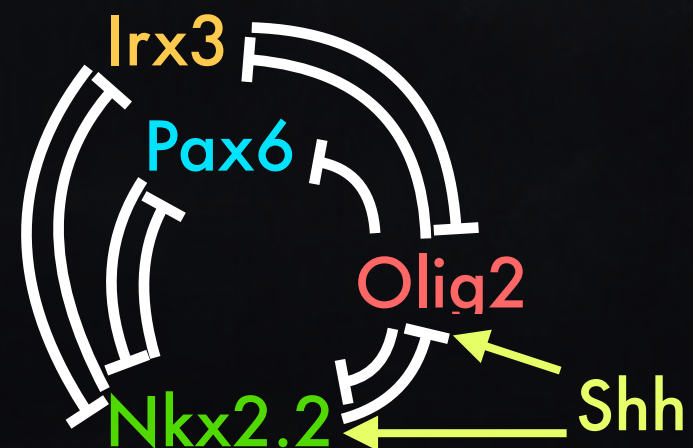
How does gene regulation control the patterning mechanism?

What do we know?

Mutant patterns



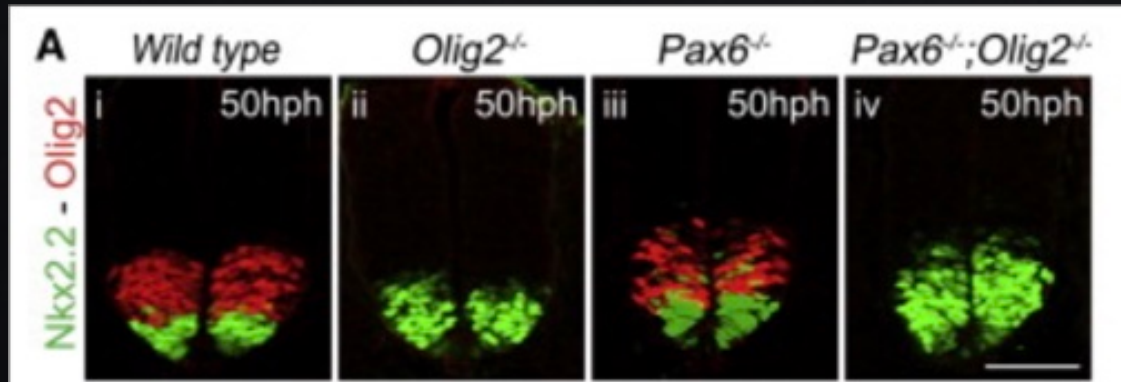
GRN topology (causality)



# Example 3: Developmental GRNs / Neural tube patterning

What do we know?

Mutant patterns



GRN topology (causality)



How do we encode it  
in a quantitative model?

Set of ODEs

Production Degradation

$$\dot{x}_I = f_I(x_N, x_O) - \delta_I x_I$$

$$\dot{x}_P = f_P(x_N, x_O) - \delta_P x_P$$

$$\dot{x}_O = f_O(x_N, x_I, S) - \delta_O x_O$$

$$\dot{x}_N = f_N(x_O, x_P, x_I, S) - \delta_N x_N$$

Question:

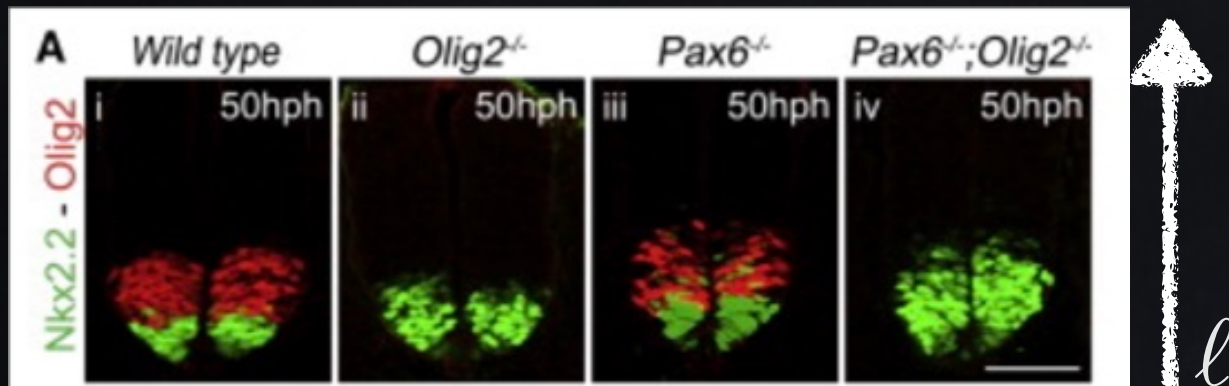
- Try to write your own  
regulatory functions  $f()$

# Example 3: Developmental GRNs / Neural tube patterning

What do we know?

How do we encode it in a quantitative model?

Mutant patterns



GRN topology (causality)



Set of ODEs

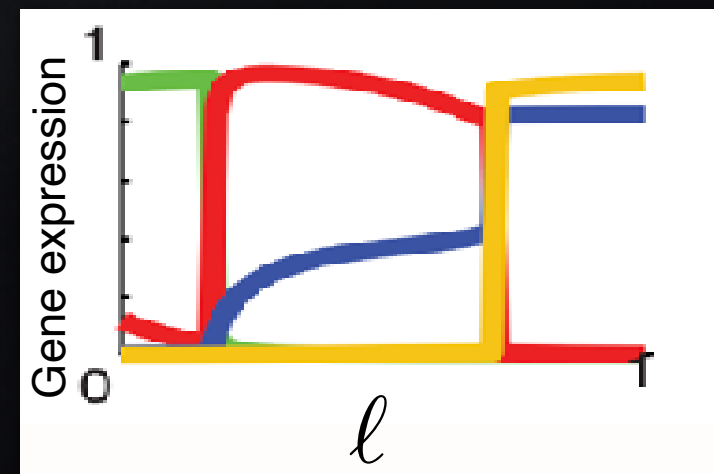
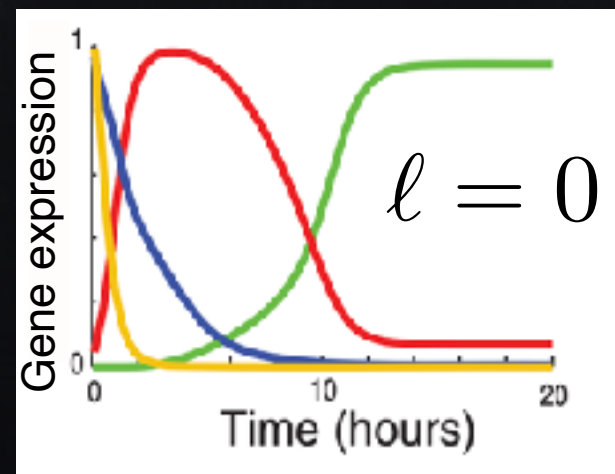
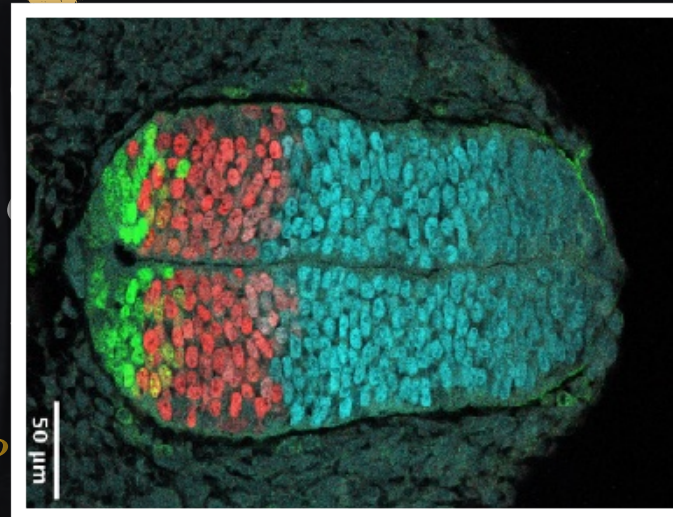
Production Degradation

$$\dot{x}_I = f_I(x_N, x_O) -$$

$$\dot{x}_P = f_P(x_N, x_O) -$$

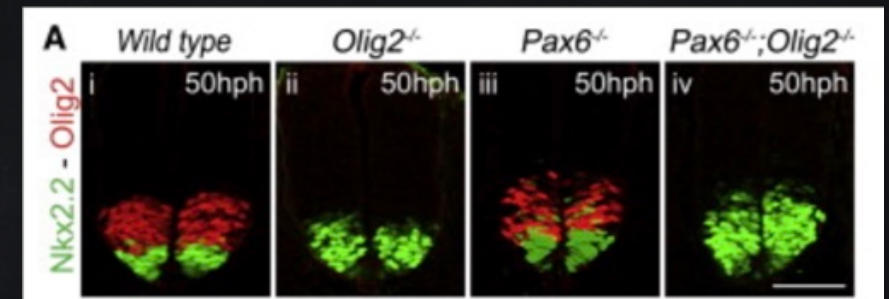
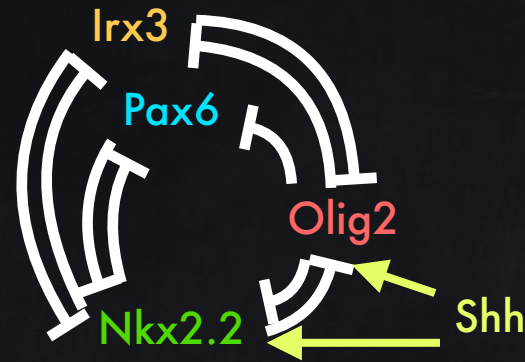
$$\dot{x}_O = f_O(x_N, x_I, S)$$

$$\dot{x}_N = f_N(x_O, x_P, x_P)$$



# Example 3: Developmental GRNs / Neural tube patterning

What do we know?



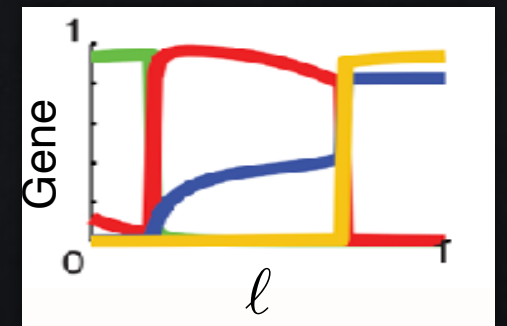
How do we encode it in a quantitative model?

$$\dot{x}_I = f_I(x_N, x_O) - \delta_I x_I$$

$$\dot{x}_P = f_P(x_N, x_O) - \delta_P x_P$$

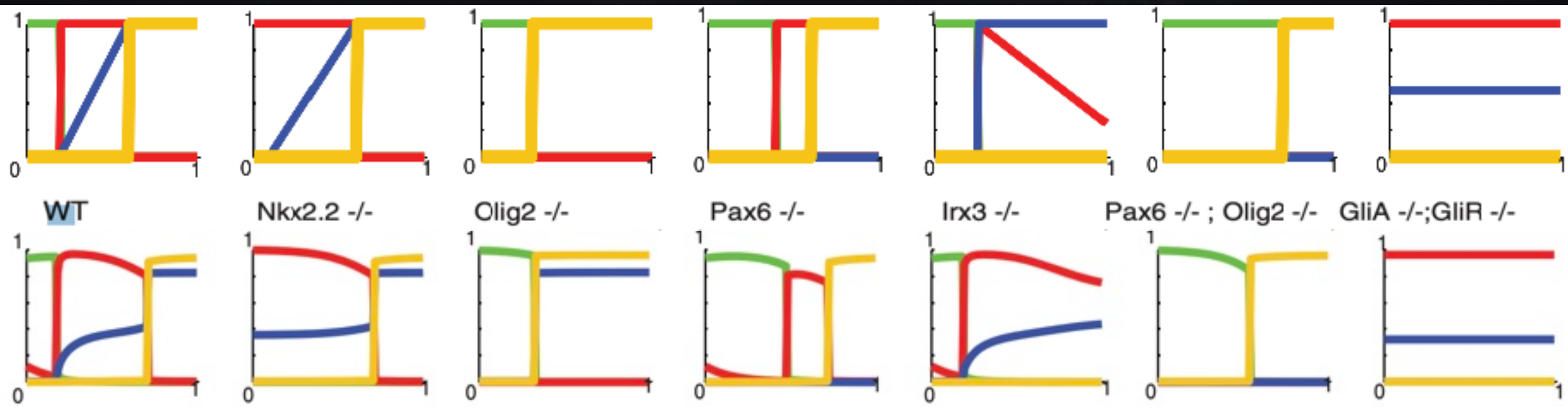
$$\dot{x}_O = f_O(x_N, x_I, S) - \delta_O x_O$$

$$\dot{x}_N = f_N(x_O, x_P, x_I, S) - \delta_N x_N$$



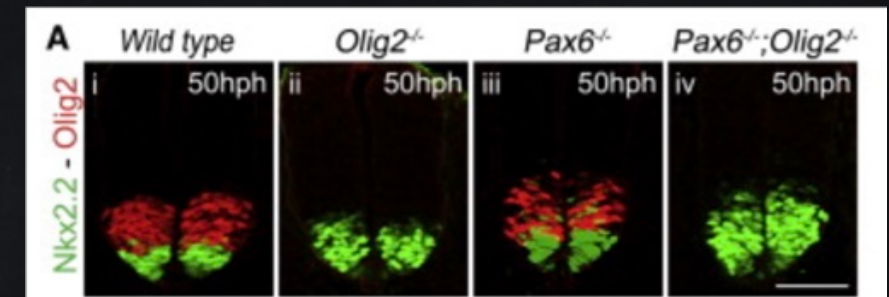
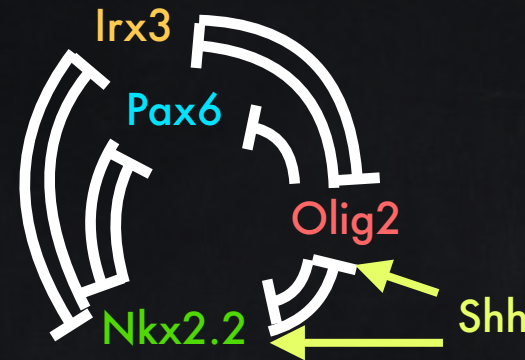
How do I compare the data with the model? (Likelihood)?

Target  
(Exp.)



# Example 3: Developmental GRNs / Neural tube patterning

What do we know?



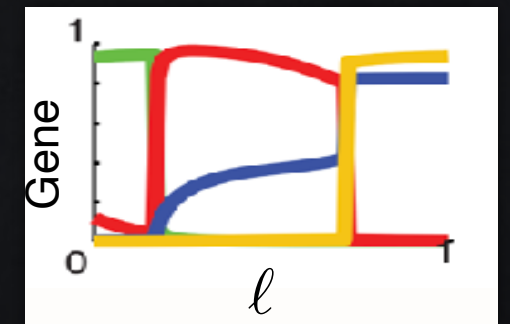
How do we encode it in a quantitative model?

$$\dot{x}_I = f_I(x_N, x_O) - \delta_I x_I$$

$$\dot{x}_P = f_P(x_N, x_O) - \delta_P x_P$$

$$\dot{x}_O = f_O(x_N, x_I, S) - \delta_O x_O$$

$$\dot{x}_N = f_N(x_O, x_P, x_I, S) - \delta_N x_N$$



How do I compare the data with the model? (Likelihood)?

Target  
(Exp.)

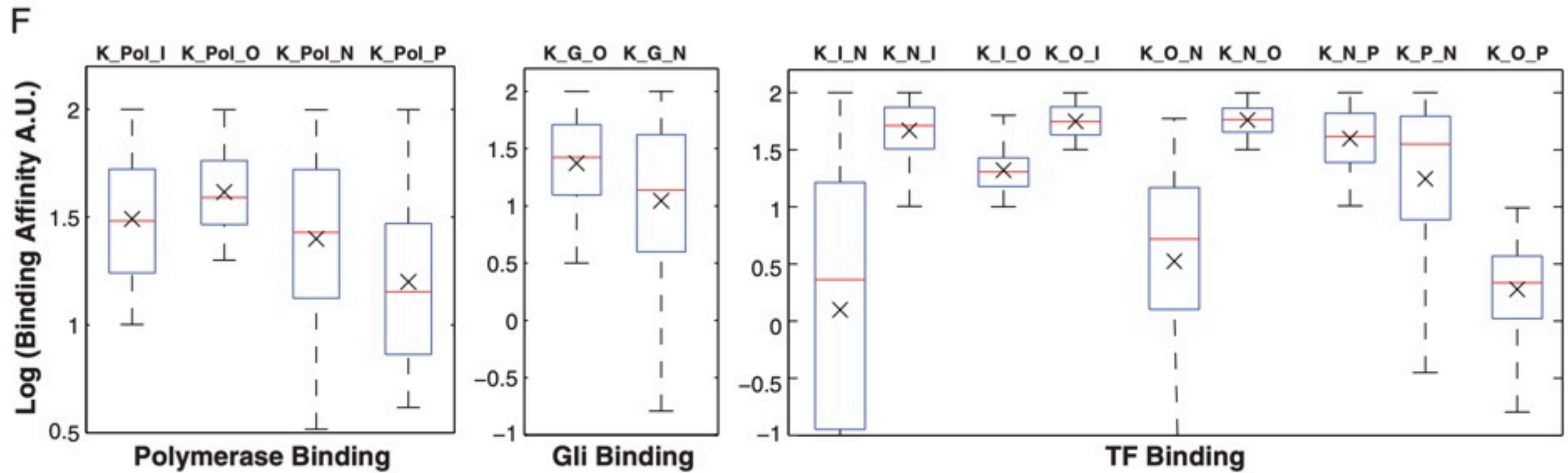
Model



Question:

- Try to write your own distance function

# Example 3: Developmental GRNs / Neural tube patterning



© 2014. Published by The Company of Biologists Ltd | Development (2014) 141, 3868-3878 doi:10.1242/dev.112573

THE COMPANY OF  
Biologists

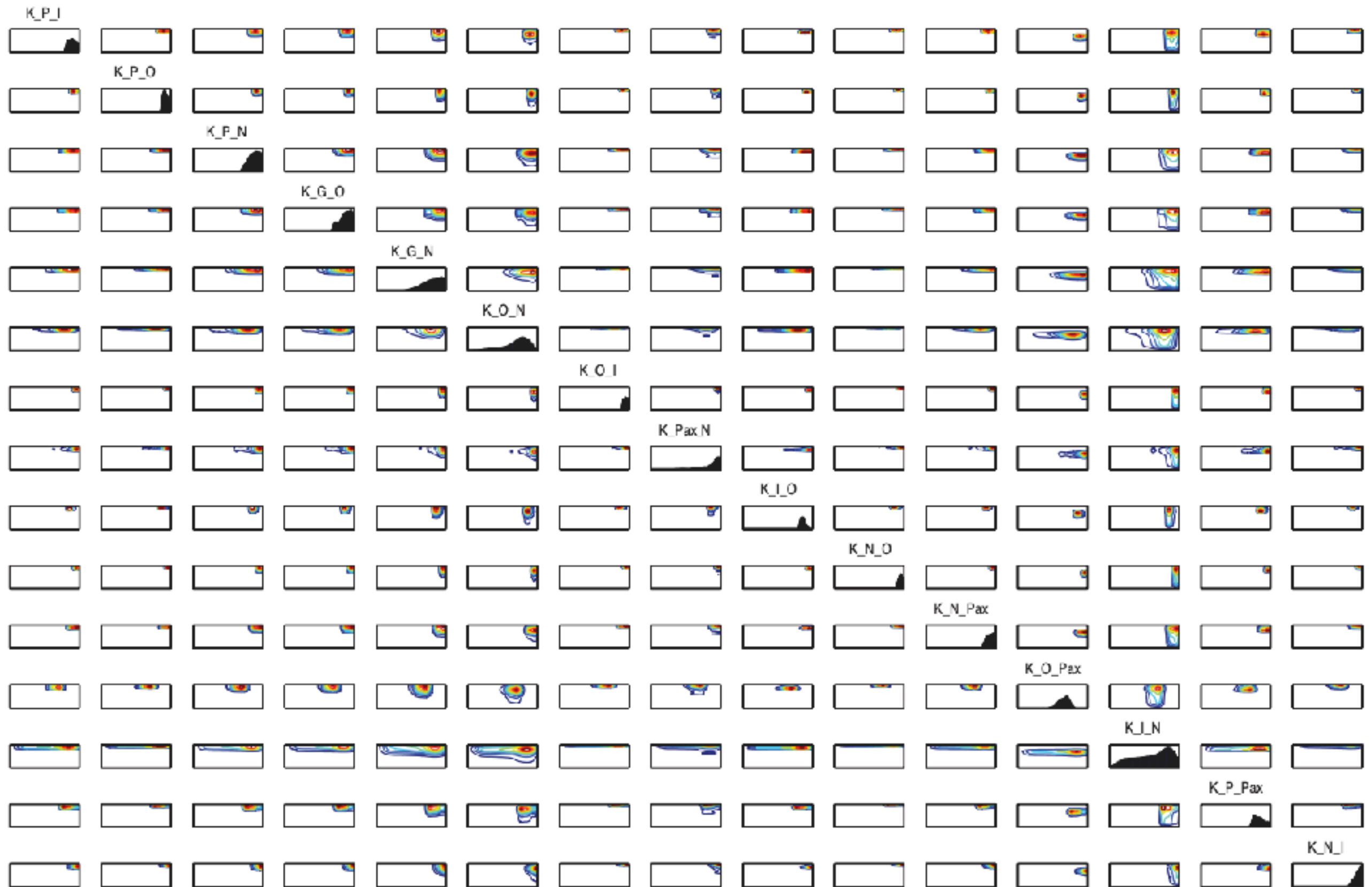
## RESEARCH ARTICLE

### A theoretical framework for the regulation of Shh morphogen-controlled gene expression

Michael Cohen<sup>1</sup>, Karen M. Page<sup>2</sup>, Ruben Perez-Carrasco<sup>2</sup>, Chris P. Barnes<sup>3</sup> and James Briscoe<sup>1,\*</sup>

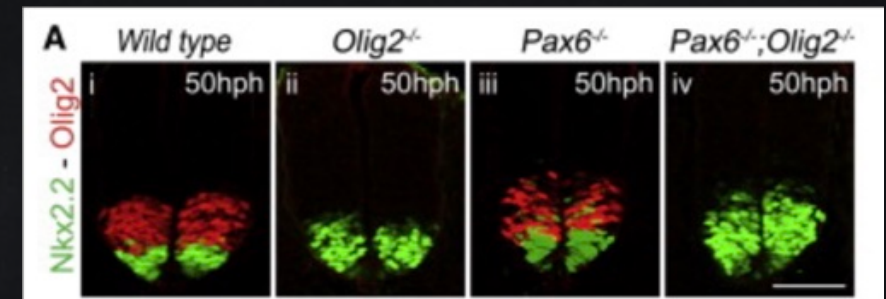
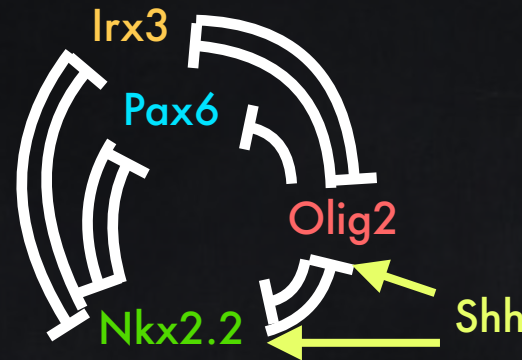
The affinity threshold model came from experiments in the early

# Example 3: Developmental GRNs / Neural tube patterning



# Example 3: Developmental GRNs / Neural tube patterning

What do we know?



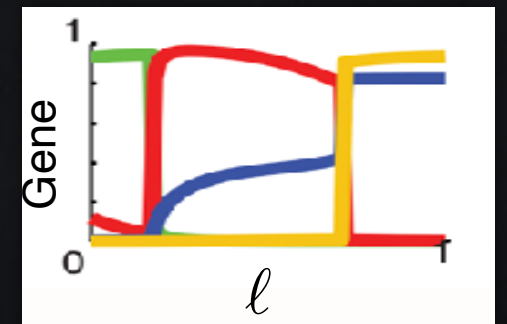
How do we encode it in a quantitative model?

$$\dot{x}_I = f_I(x_N, x_O) - \delta_I x_I$$

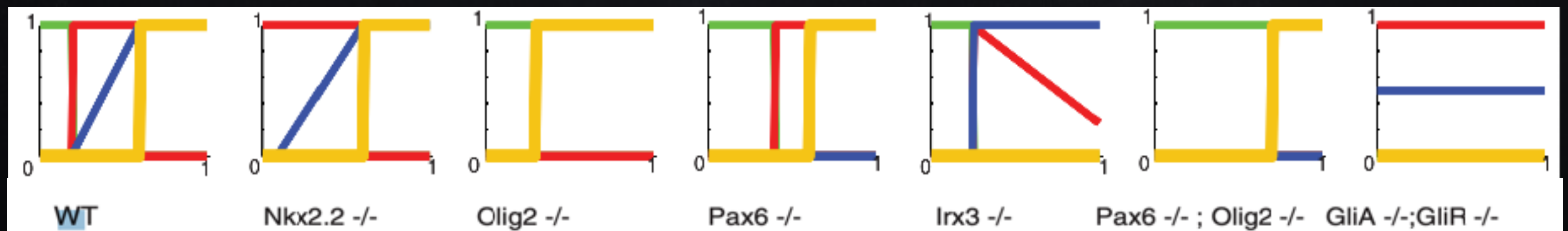
$$\dot{x}_P = f_P(x_N, x_O) - \delta_P x_P$$

$$\dot{x}_O = f_O(x_N, x_I, S) - \delta_O x_O$$

$$\dot{x}_N = f_N(x_O, x_P, x_I, S) - \delta_N x_N$$



How do I compare the data with the model (Likelihood)?



Results in a credible set of parameters that encode the network and can be used to test hypotheses in the future!

# Beyond Example 3: Developmental GRNs / Neural tube patterning

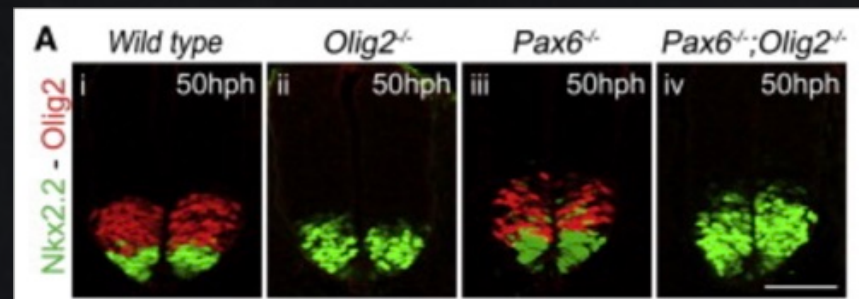
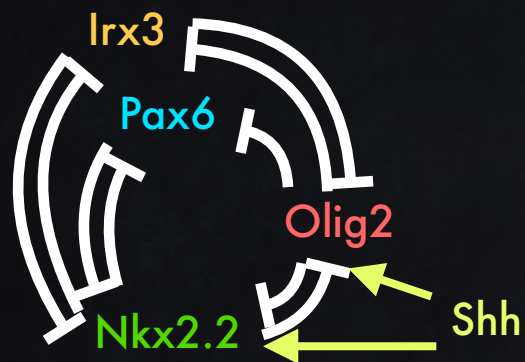


Human  
30 days



Mouse  
9 days

# Beyond Example 3: Developmental GRNs / Neural tube patterning



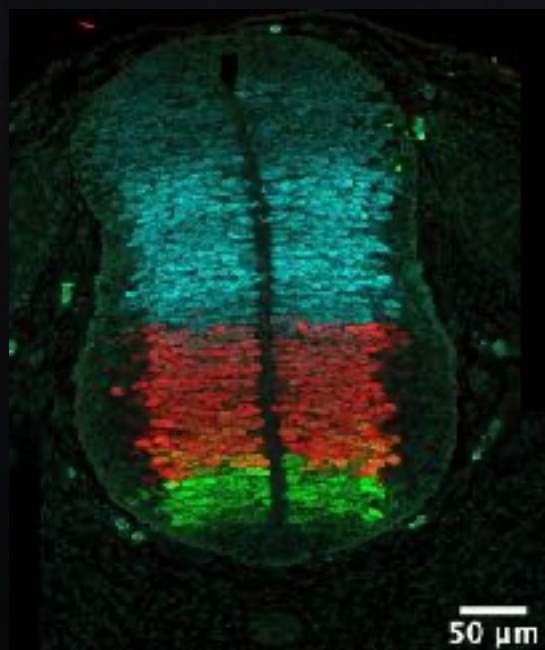
$$\dot{x}_I = f_I(x_N, x_O) - \delta_I x_I$$

$$\dot{x}_P = f_P(x_N, x_O) - \delta_P x_P$$

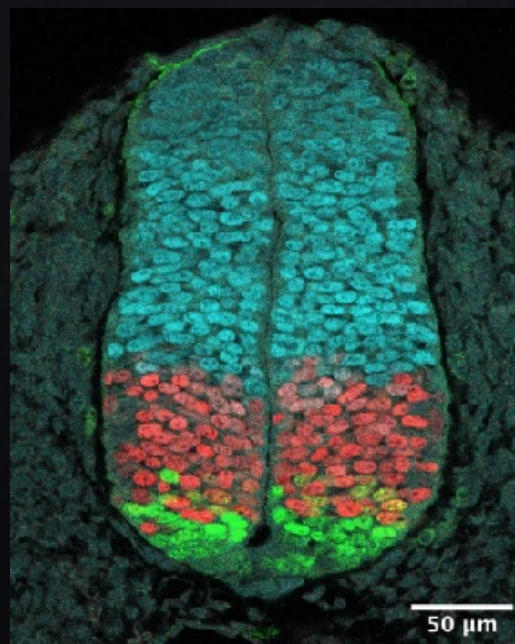
$$\dot{x}_O = f_O(x_N, x_P, S) - \delta_O x_O$$

$$\dot{x}_N = f_N(x_O, x_P, x_P, S) - \delta_1 x_N$$

Results in a credible set of parameters that encode the network and can be used to test hypotheses in the future!

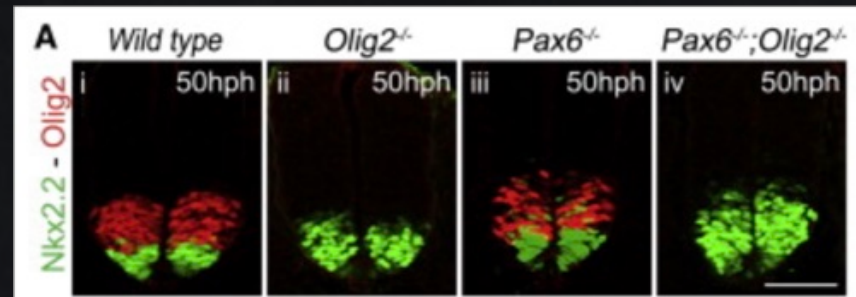
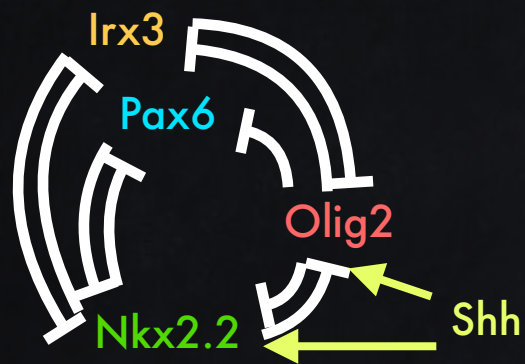


human



mouse

# Beyond Example 3: Developmental GRNs / Neural tube patterning



$$\dot{x}_I = f_I(x_N, x_O) - \delta_I x_I$$

$$\dot{x}_P = f_P(x_N, x_O) - \delta_P x_P$$

$$\dot{x}_O = f_O(x_N, x_P, S) - \delta_O x_O$$

$$\dot{x}_N = f_N(x_O, x_P, x_P, S) - \delta_1 x_N$$

Results in a credible set of parameters that encode the network and can be used to test hypotheses in the future!

## RESEARCH ARTICLE

### Species-specific pace of development is associated with differences in protein stability

Teressa Rayon<sup>1,\*</sup>, Despina Stamatakis<sup>1,†</sup>, Ruben Perez-Carrasco<sup>1,2,3,†</sup>, Lorena Garcia-Perez<sup>1</sup>, Christopher Barrington...

+ See all authors and affiliations

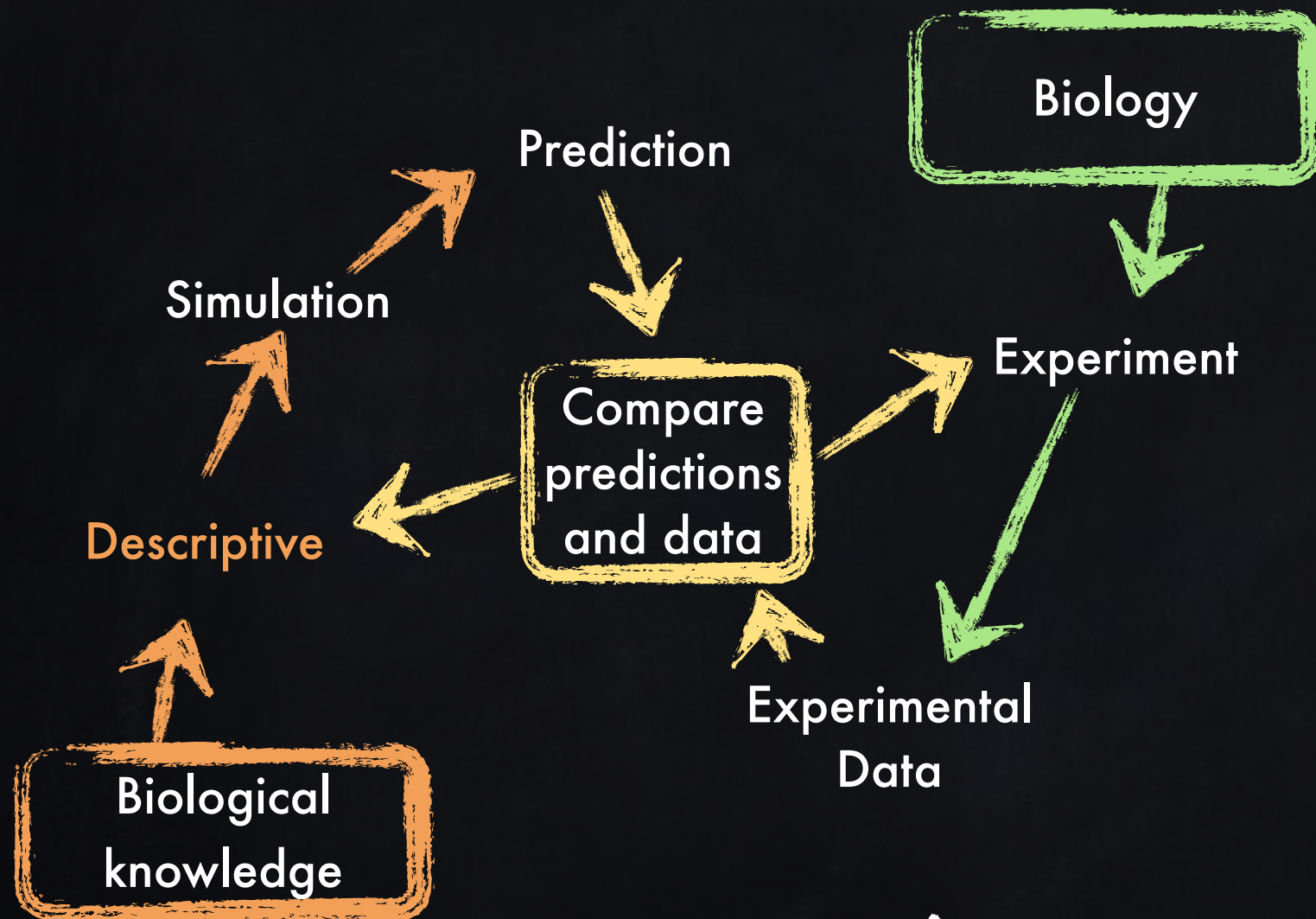
Science 10 Sep 2020:  
Vol. 369, Issue 6510, eaba7667  
DOI: 10.1126/science.aba7667

Which? Where?

How?

When?

# Summary



$$\frac{P(\delta)P(data|\delta)}{P(data)} = P(\delta|data)$$

